Notes for the Master's course

Sheaves in Topology

Taught at Utrecht University
by Dr Remy van Dobben de Bruyn

Notes by Ben Mason, Marcel Masqué Salgado,
and Splinter Suidman

$$\mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

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Introduction

These notes are based on lectures given by Remy van Dobben de Bruyn for the Master's course *Sheaves in Topology*, taught at Utrecht University in the spring semester of 2023–2024.[1]

The prerequisites for this course are a solid understanding of point-set topology, basic knowledge of fundamental groups and covering spaces, familiarity with the language of categories, and a working knowledge of modules over rings.

Recommended literature Standard works:

- Iversen, *Cohomology of sheaves* [Ive86]
- Bredon, *Sheaf theory* [Bre97]
- Tennison, *Sheaf theory* [Ten75]
- Kashiwara and Schapira, *Sheaves on manifolds* [KS94]
- Stacks project authors, *The Stacks project* [Sta24, Chapter 006A] (chapter on sheaves)

More advanced texts:

- Dimca, *Sheaves in topology* [Dim04]
- Mac Lane and Moerdijk, *Sheaves in geometry and logic* [MM94]

Exodromy correspondence (research papers):

- Treumann, 'Exit paths and constructible stacks' [Tre09]
- Curry and Patel, 'Classification of constructible cosheaves' [CP20]

course content

[1] See https://cursusplanner.uu.nl/course/WISM501/2023/SEM2 for the course description.

LECTURE 1

Motivation, sheaves and presheaves

1.1 Introduction

Not sure if we want to add the examples etc.

1.2 Sheaves and presheaves

Definition 1.2.1 · Let X be a topological space. Write $\mathrm{Open}(X)$ for the partially ordered set of opens of X. A *presheaf* of sets on X is a functor

$$F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Set}.$$

By changing the codomain we can obtain, for example, presheaves of abelian groups. In this course, we will focus almost entirely on presheaves of sets and of abelian groups. Let $U \subseteq V$ be open sets of X. The inclusion under F gives a *restriction* map $F(V) \to F(U)$. The naming comes from the following example:

Example 1.2.2 · Let X and Z be topological spaces. The assignment $h_Z : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Set}$ by $U \mapsto \mathbf{Top}(U, Z)$ can be turned into a presheaf: given $U \subseteq V$ opens of X, define

$$r_{UV} : \mathbf{Top}(V, Z) \to \mathbf{Top}(U, Z)$$

by $f \mapsto f|_U$.

We generalise the notation of function restriction. For $F(V) \to F(U)$, we denote the map pointwise by $s \mapsto s|_U$.

Definition 1.2.3 · Let X be a topological space and $\mathcal{F}$ a presheaf

$$\mathcal{F} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Set}.$$

We call $\mathcal{F}$ a *sheaf* if it satisfies the *sheaf condition*: for every open $U \subseteq X$ and every open cover $(U_i)_{i \in I}$ of U with $\bigcup_i U_i = U$, the map

$$\mathcal{F}(U) \to \prod_{i \in I} \mathcal{F}(U_i), \quad s \mapsto (s|_{U_i})_{i \in I}$$

has the following two properties:

(i) it is injective, and
(ii) its image satisfies a *gluing condition*: it is given by $\{ (s_i)_{i \in I} \mid s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \forall i, j \in I \}$.

Remark 1.2.4 · One checks that the sheaf condition equivalent to requiring that

$$\mathcal{F}(U) \longrightarrow \textstyle\prod_i \mathcal{F}(U_i) \underset{\beta}{\overset{\alpha}{\rightrightarrows}} \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

is an equaliser diagram for all U open in X and for all $(U_i)_{i \in I}$ open covers of U , where $\alpha : (s_i)_{i \in I} \mapsto s_i|_{U_i \cap U_j}$ and $\beta : (s_i)_{i \in I} \mapsto s_j|_{U_i \cap U_j}$.

Lemma 1.2.5 · *The presheaf h_Z from Example 1.2.2 is a sheaf.*

Proof. If two functions agree on every open of a cover of U they agree on U, this gives Definition 1.2.3(i). For (ii), we use the pasting lemma. □

Example 1.2.6 · Let Z be a discrete topological space, let X be a topological space. Given an open subset U of X, a map $f : U \to Z$ is continuous if and only if it is locally constant. The sheaf h_Z is called the *constant sheaf* on the set Z, labelled $\underline{Z}$ or $\underline{Z}_X$. Explicitly, $\underline{Z}$ is given by

$$\underline{Z} : U \mapsto \mathbf{Top}(U, Z),$$

where Z is endowed with the discrete topology.

Example 1.2.7 · If X is a manifold, then the assignment $U \mapsto C^\infty(U, \mathbb{R})$ is a sheaf of $\mathbb{R}$ vector spaces. One can show that the assignment $U \mapsto \Omega^k(U)$ (smooth differential k-forms) is a sheaf.

Lemma 1.2.8 (sheaf of sections) · *Let $f : Y \to X$ be a continuous map of topological spaces. The assignment on opens of X given by*

$$h_{Y/X} : U \mapsto \{ s : U \to f^{-1}(U) \mid f \circ s = \mathrm{id}_U \} =: \mathbf{Top}_{/X}(U, Y)$$

is a sheaf.

Proof. One can prove the above lemma in a similar way we proved Lemma 1.2.5. Alternatively, consider the diagram

$$\begin{tikzcd}
\mathbf{Top}_{/X}(U,Y) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathbf{Top}(U,Y) \arrow[d, "f\circ -"] \\
\{U \to X\} \arrow[r] & \mathbf{Top}(U,X)
\end{tikzcd}$$

and check that it is a pullback. We will come back to this in more detail in later lectures. □

The sheaf $h_{Y/X}$ of the lemma is called the *sheaf of sections*. The example in the lemma above is why the elements of $\mathcal{F}(U)$ for an arbitrary sheaf $\mathcal{F}$ are more generally also called *sections*.

Example 1.2.9 · Let $Y \xrightarrow{f} X$ be the two-to-one cover of the circle: $f : z \mapsto z^2$, with $X := S^1$ and $Y := S^1$. On small intervals $U \subseteq X$ we get $f^{-1}(U) \cong U \times \{1, 2\}$. We thus have two sections: $U \mapsto (U, 1)$ and $U \mapsto (U, 2)$. So $h_{Y/X}(U)$ has two elements On V a union of small intervals, we get $2^{|\pi_0(V)|}$ elements, where $|\pi_0(V)|$ is the number of path components of V. On $W = X$, we get no sections. If $s : X \to Y$ is a section, then the induced map $s_* : \pi_1(X) \to \pi_1(Y)$ is a section to the map $f_* : \pi_1(Y) \to \pi_1(X)$. But this induced map is multiplication by 2, and it does not have a section.

LECTURE 2

Morphisms of (pre)sheaves, Yoneda lemma, étalé space

'You can do what you want, it's a free world (on one generator).'

2.1 Morphisms of (pre)sheaves

Definition 2.1.1 · Let $\mathscr{C}$ be a small category. Then $\mathbf{PSh}(\mathscr{C})$ is the functor category $\mathbf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathbf{Set})$: its objects are functors $F : \mathscr{C}^{\mathrm{op}} \to \mathbf{Set}$, and its morphisms $\alpha : F \Rightarrow G$ are *natural transformations*, that is, collections of functions $\alpha_X : F(X) \to G(X)$ for all $X \in \mathscr{C}$ such that the diagram

$$\begin{tikzcd} F(Y) \arrow[r, "\alpha_Y"] \arrow[d, "F(f)"'] & G(Y) \arrow[d, "G(f)"] \\ F(X) \arrow[r, "\alpha_X"'] & G(X) \end{tikzcd}$$

commutes for all $f : X \to Y$ in $\mathscr{C}$.

Definition 2.1.2 · Let X be a topological space. Then the category of presheaves on X is $\mathbf{PSh}(X) := \mathbf{PSh}(\mathrm{Open}(X))\mathbf{PSh}(X)$ and the category of sheaves on X is the full subcategory $\mathbf{Sh}(X) \subseteq \mathbf{PSh}(X)\mathbf{Sh}(X)$ on the sheaves.

Lemma 2.1.3 · *Let $\mathscr{C}$ be a small category. Then:*

(i) $\mathbf{PSh}(\mathscr{C})$ *is a category.*

(ii) The category $\mathbf{PSh}(\mathscr{C})$ *has all (small) limits and colimits, and they are computed objectwise; e.g., for presheaves $F, G \in \mathbf{PSh}(\mathscr{C})$ the natural map*

$$(F \times G)(U) \to F(U) \times G(U)$$

is an isomorphism for all open $U \subseteq X$.

(iii) A natural transformation $\alpha : F \Rightarrow G$ between presheaves F and G on $\mathscr{C}$ is invertible if and only if the component $\alpha_X : F(X) \to G(X)$ is a bijection for all $X \in \mathscr{C}$.

Remark 2.1.4 · Colimits in the category $\mathbf{Sh}(X)$ of sheaves will be more complicated.

Example 2.1.5 · Recall that we defined a sheaf h_Z on X for $Z \in \mathbf{Top}$ (Example 1.2.2). If $g : Z \to Z'$ is a continuous map, then we get a natural transformation $h_Z \Rightarrow h_{Z'}$ with component

$$\mathbf{Top}(U, Z) \to \mathbf{Top}(U, Z'), \quad f \mapsto g \circ f$$

at $U \in \mathrm{Open}(X)$. One checks that this is natural.

Example 2.1.6 · For the sheaf of sections (Lemma 1.2.8), a map $g : Y \to Y'$ over X induces a natural transformation $h_{Y/X} \Rightarrow h_{Y'/X}$ with component

$$\mathbf{Top}_{/X}(U, Y) \to \mathbf{Top}_{/X}(U, Y'), \quad s \mapsto g \circ s$$

at $U \in \mathrm{Open}(X)$. This is again natural in U.

In fact, the sheaf h_Z is a special case of $h_{Y/X}$:

Lemma 2.1.7 · *If $Y = Z \times X \xrightarrow{\mathrm{pr}_X} X$ in $\mathbf{Top}_{/X}$, then the sheaves $h_{Y/X}$ and h_Z are isomorphic.* Picture

Proof. For $U \in \mathrm{Open}(X)$, define

$$\mathbf{Top}_{/X}(U, Z \times X) \to \mathbf{Top}(U, Z), \quad s \mapsto \mathrm{pr}_Z \circ s$$

and

$$\mathbf{Top}(U, Z) \to \mathbf{Top}_{/X}(U, Z \times X), \quad f \mapsto (u \mapsto (f(u), u)).$$

These maps are inverses. Both transformations are natural. For the first: given an inclusion of opens $U \subseteq V \subseteq X$, the diagram

$$\begin{tikzcd}
\mathbf{Top}_{/X}(V, Z \times X) \arrow[r, "\mathrm{pr}_Z \circ -"] \arrow[d, "(-)|_U"'] & \mathbf{Top}(V, Z) \arrow[d, "(-)|_U"] \\
\mathbf{Top}_{/X}(U, Z \times X) \arrow[r, "\mathrm{pr}_Z \circ -"'] & \mathbf{Top}(U, Z)
\end{tikzcd}$$

commutes. □

Example 2.1.8 · Let $Y \to X$ be the two-to-one cover $S^1 \to S^1$. Then $h_{Y/X}$ is not isomorphic to h_Z for any $Z \in \mathbf{Top}$ (but it is locally isomorphic to $h_{\{1,2\}}$, as we will see later). Ref back

Suppose $h_{Y/X} \cong h_Z$ for $Z \in \mathbf{Top}$. Then $Z \neq \varnothing$, for

$$h_\varnothing(U) = \begin{cases} \varnothing & \text{if } U \neq \varnothing, \\ * & \text{if } U = \varnothing. \end{cases}$$

But we have seen that $h_{Y/X}$ is nonempty for a small enough interval. So $Z \neq \varnothing$, so constant maps show that $h_Z(U) \neq \varnothing$ for all U. But last lecture, we saw that $h_{Y/X}(X) = \varnothing$. Ref

2.2 Yoneda lemma

Definition 2.2.1 · Let $\mathscr{C}$ be a small category. Then the *representable presheaf* on $X \in \mathscr{C}$ is the presheaf

$$h_X : \mathscr{C}^{\mathrm{op}} \to \mathbf{Set}, \quad Y \mapsto \mathrm{Hom}_{\mathscr{C}}(Y, X)$$

with restriction map

$$f^* : \mathrm{Hom}_{\mathscr{C}}(X, Y') \to \mathrm{Hom}_{\mathscr{C}}(X, Y), \quad g \mapsto g \circ f.$$

induced by $f : Y \to Y'$ in $\mathscr{C}$.

Remark 2.2.2 · The sheaf h_Z on X from Example 1.2.2 is *not* a representable presheaf on $\mathrm{Open}(X)$. The sheaf h_Z sends an open $U \subseteq X$ to the set $\mathbf{Top}(U, Z)$ of all continuous maps $U \to Z$, whereas the representable presheaf h_V represented by an open $V \subseteq X$ sends $U \subseteq X$ to the set $\mathrm{Hom}_{\mathrm{Open}(X)}(U, V)$ of inclusions $U \hookrightarrow V$ (a subsingleton set). The sheaf h_Z can be regarded as the *restriction* of the representable presheaf $h_Z = \mathrm{Hom}_{\mathbf{Top}}(-, Z)$ on the (non-small) category of spaces to the (non-full) subcategory $\mathrm{Open}(X) \subseteq \mathbf{Top}$.

In some sense, the representable presheaf represented by X is 'freely generated' by the section $\mathrm{id}_X \in h_X(X)$.

Lemma 2.2.3 (Yoneda lemma [Rie16, Theorem 2.2.4]) · *Let $F : \mathscr{C}^{\mathrm{op}} \to \mathbf{Set}$ be a functor, $X \in \mathscr{C}$. Then the map*

$$\Phi : \mathrm{Hom}_{\mathbf{PSh}(\mathscr{C})}(h_X, F) \to F(X), \quad \alpha \mapsto \alpha_X(\mathrm{id}_X)$$

is a bijection that is natural in F and X.

Proof. We leave the naturality in F and X as an exercise. Part of the exercise is to work out what is meant by naturality in F and X. The inverse of Φ will be a map

maybe add details

$$\Psi : F(X) \to \mathrm{Hom}_{\mathbf{PSh}(C)}(h_X, F)$$

defined by $s \mapsto (f \mapsto F(f)(s))_{Y \in \mathscr{C}}$. The first thing to check is that this map lands in $\mathrm{Hom}_{\mathbf{PSh}(\mathscr{C})}(h_X, F)$. That is, we need to check $\Psi(s)$ is a natural transformation. This is true: given $g : Y \to Z$ in $\mathscr{C}$ the diagram

$$\begin{tikzcd}
\mathrm{Hom}(Z, X) \arrow[r, "\Psi(s)_Z"] \arrow[d, "-\circ g"'] & F(Z) \arrow[d, "F(g)"] \\
\mathrm{Hom}(Y, X) \arrow[r, "\Psi(s)_Y"'] & F(Y)
\end{tikzcd}$$

commutes, since we have $F(g)(\Psi(s)_Z(f)) = F(g)(F(f)(s)) = F(fg)(s) = \Psi(s)_Y(fg)$. We check that Ψ provides an inverse for Φ. One side is immediate: $\Phi\Psi(s) = \Psi(s)_X(\mathrm{id}_X) = F(\mathrm{id}_X)(s) = s$. For the converse, let $\alpha : h_X \to F$ be a natural transformation. We check that $\Psi\Phi(\alpha) = \alpha$. Let $Y \in \mathscr{C}$, and $f \in \mathrm{Hom}(Y, X)$. The naturality of α gives a commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}(X, X) \arrow[r, "\alpha_X"] \arrow[d, "-\circ f"'] & F(X) \arrow[d, "F(f)"] \\
\mathrm{Hom}(Y, X) \arrow[r, "\alpha_Y"'] & F(Y)
\end{tikzcd}$$

We now have $\Psi(\Phi\alpha)_Y(f) = F(f)(\Phi\alpha) = F(f)(\alpha_X(\mathrm{id}_X)) = \alpha_Y(\mathrm{id}_X \circ f) = \alpha_Y(f)$ by plugging in id_X into the diagram above. □

2.3 Étalé space

In the coming lectures, we will show that every sheaf on a space X is isomorphic to $h_{Y/X}$ for some map $Y \to X$. We will construct a functor $\mathrm{sp} : \mathbf{PSh}(X) \to \mathbf{Top}_{/X}$ which sends a presheaf to its *étalé space* (also called *espace étalé*, (wrongly) *étale space* or *sheaf space* (but we do not like that)).

The idea is as follows: a presheaf F is determined by its values $F(U)$ for opens $U \subseteq X$ and the restriction maps. By the Yoneda lemma, the values can be written as $F(U) \cong \mathrm{Hom}_{\mathbf{PSh}(X)}(h_U, F)$. For h_U, the étalé space will be the space $U \to X$ over X. In general, a presheaf F is 'generated by copies of h_U modulo relations from the restrictions'. We define $\mathrm{sp}(F) \to X$ by glueing copies of U 'along the same colimit diagram' (but now in $\mathbf{Top}_{/X}$ instead of $\mathbf{PSh}(X)$).

Definition 2.3.1 · Let F be a presheaf on X. The *étalé space* $\mathrm{sp}(F)$ is the quotient of the coproduct

$$\coprod_{U \in \mathrm{Open}(X)} F(U) \times U$$

(where $F(U)$ is endowed with the discrete topology), with notation $(s, x)_U$ for the point given by $s \in F(U)$ and $x \in U$ in the factor indexed by U, by the equivalence relation generated by

$$(s|_U, x)_U \sim (s, x)_V$$

for $(s, x) \in F(V) \times U$ for $x \in U \subseteq V$.

Categorically, the étalé space is the coequaliser

$$\mathrm{sp}(F) = \mathrm{coeq}\Big(\coprod_{U \subseteq V \subseteq X} F(V) \times U \rightrightarrows \coprod_{U \subseteq X} F(U) \times U\Big)$$

where the arrows are given by

$$\begin{array}{ccccc} F(U) \times U & \xleftarrow{\mathrm{incl}^* \times \mathrm{id}} & F(V) \times U & \xrightarrow{\mathrm{id} \times \mathrm{incl}} & F(V) \times V \\ \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ (s|_U, x) & \longleftarrow\!\shortmid & (s, x) & \longmapsto & (s, x) \end{array}$$

where $\mathrm{incl} : U \hookrightarrow V$ denotes the inclusion map.

Alternatively, the étalé space $\mathrm{sp}(F)$ is the coend

$$\mathrm{sp}(F) = \int^{U \in \mathrm{Open}(X)} F(U) \times U.$$

To understand the étalé space a bit better, we have to understand the equivalence relation generated by the given relation, with respect to which we take the quotient.

Remark 2.3.2 · Let $\sim$ be a relation on a set X. The equivalence relation $\approx$ on X generated by $\sim$ can be explicitly defined as follows: we say $x \approx y$ if and only if there is a finite zigzag

$$\begin{tikzcd} & a_1 & & a_3 & & a_{2n-1} & \\ x = a_0 \arrow[ur, "\sim"] & & a_2 \arrow[ul, "\sim"'] \arrow[ur, "\sim"] & & \dots \arrow[ul, "\sim"'] \arrow[ur, "\sim"] & & a_{2n} = y \arrow[ul, "\sim"'] \end{tikzcd}$$

of elements $a_i \in X$ such that $x = a_0$, $y = a_{2n}$, $a_{2i} \sim a_{2i+1}$ for $0 \leqslant i < n$ and $a_{2i} \sim a_{2i-1}$ for $0 < i \leqslant n$.

Example 2.3.3 · To compute the coequaliser of the diagram of sets

$$\{1, 2\} \underset{-\cdot 2}{\overset{-\cdot 1}{\rightrightarrows}} \{1, 2, 3, 4\}$$

where the maps multiply by one and two, respectively, we need to understand the equivalence relation generated by $x \sim 2x$ for all $x \in \{1, 2\}$. That is, we have $1 \sim 2$ and $2 \sim 4$, and nothing more. The equivalence classes in the relation $\approx$ generated by $\sim$ then are $\{1, 2, 4\}$ and $\{3\}$.

Lemma 2.3.4 · *If $(s, x)_U \in F(U) \times U$ and $(t, y)_V \in F(V) \times V$, then $(s, x)_U \approx (t, y)_V$ (that is, they are equivalent under the equivalence relation generated by $\sim$ as in Definition 2.3.1) if and only if $x = y$ and there exists an open neighbourhood $W \subseteq U \cap V$ of x such that $s|_W = t|_W$.*

One can prove this lemma using the explicit description of $\approx$ given in Remark 2.3.2, but this is difficult combinatorics! A better proof shows that the relation on the right in the equivalence of the lemma is an equivalence relation and that the former statement implies the latter (i.e., '$\sim$ implies $\approx$').

Example 2.3.5 · If $F = h_U$ is the representable presheaf on X represented by an open $U \subseteq X$ (see Figure 2.1), then

$$F(V) = \mathrm{Hom}_{\mathrm{Open}(X)}(V, U) = \begin{cases} * & \text{if } V \subseteq U, \\ \varnothing & \text{if } V \not\subseteq U. \end{cases}$$

The étalé space $\mathrm{sp}(F)$ is the quotient of

$$\coprod_{V \subseteq X} F(V) \times V = \coprod_{V \subseteq U} V$$

by the equivalence relation generated by $(*, x)_V \sim (*, x)_{V'}$ for all $V, V' \subseteq U$. This just leaves U, so $\mathrm{sp}(h_U) = U$.

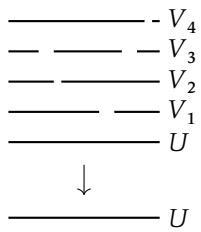

Figure 2.1 · The presheaf $F = h_U$ represented by $U \subseteq X$ of Example 2.3.5.

Remark 2.3.6 · In general, the étalé space $\mathrm{sp}(F)$ is not 'computable'. It is, however, when F is *constructible*, which we will see in one of the final lectures. Ref back

Proposition 2.3.7 · *The construction of the étalé space defines a functor* $\mathbf{Sh}(X) \to \mathbf{Top}_{/X}$.

Proof. We can cheat and use the unproven remark that the étalé space is a coend. □ prove?

LECTURE 3

Local homeomorphisms, sheaf/space adjunction

3.1 Local homeomorphisms

'I don't like annoying stuff, so let's do it the non-annoying way.'

Definition 3.1.1 · A continuous function $f : Y \to X$ is said to be a *local homeomorphism* if for all $y \in Y$ there exists an open neighbourhood $V_y \subseteq Y$ such that $f|_{V_y} : V_y \to f(V)$ is a homeomorphism onto an open subset of X.

Later we will see that we may equivalently define an étalé space over X as a topological space Y together with a local homeomorphism $Y \to X$.

Remark 3.1.2 · A local homeomorphism $f : Y \to X$ is always open. Namely let $W \subseteq Y$ be open, then it can be covered by opens of the form $W \cap V_y$, where $f|_{V_y} : V_y \to f(V_y)$ are homeomorphisms. Consequently,

$$f(W) = f(\bigcup_{y \in Y} V_y \cap W) = \bigcup_{y \in Y} f(V_y \cap W).$$

Therefore, since $f(V_y \cap W) = f|_{V_y}(V_y \cap W)$ is open in $f(V_y)$, it is open in X, thus $f(W)$ is open.

Lemma 3.1.3 · *Let*

$$\begin{tikzcd} Z \arrow[rr, "g"] \arrow[dr, "h"'] & & Y \arrow[dl, "f"] \\ & X & \end{tikzcd}$$

be a commutative triangle in **Top**. *Then,*

(i) If f and g are local homeomorphisms then so is h.
(ii) If f and h are local homeomorphisms then so is g.
(iii) If g is open and surjective and h is a local homeomorphism then so are f and g.

Proof. Argue locally. □

In particular, by (i) of the above lemma, we can show that the local homeomorphisms over X form a category $\mathbf{LocalHomeo}_{/X}$, either as a full subcategory of $\mathbf{Top}_{/X}$ by (ii), or as the slice over X of the non-full subcategory **LocalHomeo** of **Top** where all maps are local homeomorphisms.

Example 3.1.4 · Let $X, Y \in \mathbf{Top}$ and $U \subseteq X$ open. We have the following list of examples:

(i) The inclusion $U \hookrightarrow X$ is a local homeomorphism.

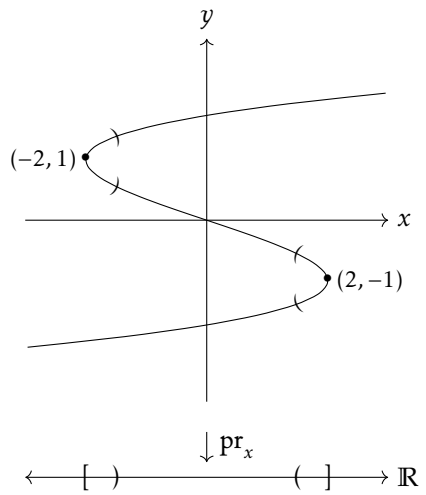

Figure 3.1 · The projection $\mathrm{pr}_x : V \to \mathbb{R}$ is not a local homeomorphism around the points $(\pm 2, \mp 1)$ since any open neighborhood is projected onto a non-open subset of $\mathbb{R}$.

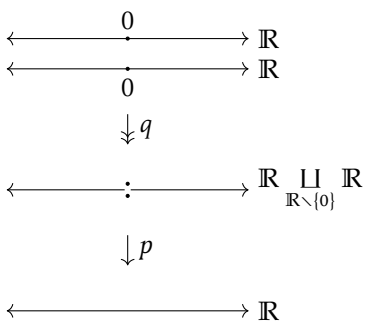

Figure 3.2 · Two local homeomorphisms related to the line with two origins

(ii) Certain covering spaces $p : Y \to X$ are local homeomorphisms. work out

(iii) Consider the zero locus $V \subseteq \mathbb{R}^2$ of the polynomial $x - y^3 - 3y$. The projection on the x-axis $\mathrm{pr}_x : V \to \mathbb{R}$ is not a local homeomorphism as shown in Figure 3.1. However, after restricting to $V \setminus \{(\pm 2, \mp 1)\}$ it is.

(iv) Consider the line with two origins $\mathbb{R} \sqcup_{\mathbb{R}\setminus\{0\}} \mathbb{R}$ obtained by gluing two copies of the real line $\mathbb{R}$ along $\mathbb{R} \setminus \{0\}$; see Figure 3.2. There is a map q from the disjoint union $\mathbb{R} \sqcup \mathbb{R}$ to the line with two origins which sends the origin in the first copy of $\mathbb{R}$ to the one of the two origins in $\mathbb{R} \sqcup_{\mathbb{R}\setminus\{0\}} \mathbb{R}$ and the origin in the other copy to the other of the two origins. There is a further map p down to $\mathbb{R}$ which collapses the two origins. Both maps p and q are local homeomorphisms.

The line with two origins is not Hausdorff. In the second assignment, we will define a sheaf on $\mathbb{R}$ whose étalé space is the line with two origins, illustrating that the étalé space is usually not Hausdorff.

Lemma 3.1.5 · *Let F be a presheaf on X. Then the maps*

$$\coprod_{U \in \mathrm{Open}(X)} F(U) \times U \xrightarrow{q} \mathrm{sp}(F) \xrightarrow{p} X$$

are local homeomorphisms, where q is the quotient map and $p : [s, x]_U \mapsto x$.

Proof. Clearly, $p \circ q$ is a local homeomorphism. Thus, it suffices to show q is open on opens of the form $(\{s|_V\} \times V)_V$, where $s \in F(U)$, $V \subseteq U$ opens. For if $q((\{s|_V\} \times V)_V)$ is open, since $(s,x)_U \sim (s|_V, x)_V$ for all $x \in V$, $q((\{s\} \times V)_U) = q((\{s|_V\} \times V)_V)$ is open, so then q is open on all basis opens. Yet since q is a quotient map, this is equivalent to $O := q^{-1} \circ q((\{s\} \times U)_U)$ being open. So let $(t,x)_V \in O$, then $x \in U$ and $(t,x)_V \sim (s,x)_U$. Hence there exists a $W \subseteq U \cap V$ such that $t|_W = s|_W$. Thus $(\{t\} \times W)_V \subseteq F(V) \times V$ is an open neighbourhood of $(t,x)_V$ in O. So q is open, therefore, by the previous lemma, p and q are local homeomorphisms. □

Of note, (iii) of Lemma 3.1.3 would not hold if we were to require g to be a quotient map instead of open. As a counterexample, let $I := [-1,1] \subseteq \mathbb{R}$. Let $Z = I \amalg I$ be the disjoint union (with coordinates $(x, \pm 1)$), $Y = Z/\{(\pm 1, -1) \sim (\pm 1, 1)\} \cong S^1$ and $X = I$. Then while g is a quotient map and h a local homeomorphism, f is not a local homeomorphism (in the points $[(\pm 1, 1)]$).

3.2 Sheaf/space adjunction

'Let's go!'

We constructed functors

$$\begin{tikzcd} \mathbf{PSh}(X) \arrow[r, shift left, "\mathrm{sp}"] & \mathbf{Top}_{/X} \arrow[l, shift left, "h_{-/X}"] \end{tikzcd}$$

and showed that sp lands in $\mathbf{LocalHomeo}_{/X}$ and $h_{-/X}$ lands in $\mathbf{Sh}(X)$.

Theorem 3.2.1 · *There is an adjunction*

$$\begin{tikzcd} \mathbf{PSh}(X) \arrow[r, shift left, "\mathrm{sp}"] \arrow[r, phantom, "\perp"] & \mathbf{Top}_{/X} \arrow[l, shift left, "h_{-/X}"] \end{tikzcd}$$

which restricts to an adjoint equivalence

$$\begin{tikzcd} \mathbf{Sh}(X) \arrow[r, shift left, "\mathrm{sp}"] \arrow[r, phantom, "\simeq"] & \mathbf{LocalHomeo}_{/X} \arrow[l, shift left, "h_{-/X}"] \end{tikzcd}$$

Example 3.2.2 · Let's check the adjunction for the sheaf $h_{U/X} = h_U = \mathrm{Hom}_{\mathrm{Open}(X)}(-, U)$. We showed in Example 2.3.5 that $\mathrm{sp}(h_{U/X}) = U$ in $\mathbf{Top}_{/X}$. On the other hand, the Yoneda Lemma 2.2.3 gives

$$\mathrm{Hom}_{\mathbf{PSh}(X)}(h_U, h_{Y/X}) \cong h_{Y/X}(U) = \mathbf{Top}_{/X}(U, Y).$$

Some formal nonsense if you want to feel fancy at dinner parties: sp is 'by definition' the left Kan extension of

$$\mathrm{Open}(X) \to \mathbf{Top}_{/X}, \quad U \mapsto (U \hookrightarrow X)$$

along the Yoneda embedding $h : \mathrm{Open}(X) \to \mathbf{PSh}(X)$.

Proof (of the adjunction in Theorem 3.2.1). Since $\mathrm{sp}(F)$ is the coequaliser of the diagram

$$\begin{tikzcd} \coprod_{U \subseteq V} F(V) \times U \arrow[r, shift left, "a"] \arrow[r, shift right, "b"'] & \coprod_U F(U) \times U \end{tikzcd}$$

we have

$$\mathbf{Top}_{/X}(\mathrm{sp}(F), Y) \cong \{ f : \coprod_U F(U) \times U \to Y \mid fa = fb \}$$

and likewise

$$\begin{aligned}\mathbf{Top}_{/X}\Big(\coprod_{U\subseteq X} F(U)\times U, Y\Big) &\cong \prod_{U\subseteq X} \mathbf{Top}_{/X}(F(U)\times U, Y)\\ &\cong \prod_{U\subseteq X} \mathrm{Map}(F(U), \mathbf{Top}_{/X}(U,Y))\\ &= \prod_{U\subseteq X} \mathrm{Map}(F(U), h_{Y/X}(U))\end{aligned}$$

and

$$\mathbf{Top}_{/X}\Big(\coprod_{U\subseteq V} F(V)\times U, Y\Big) \cong \prod_{U\subseteq V} \mathrm{Map}(F(V), h_{Y/X}(U)).$$

The maps $-\circ a, -\circ b : \prod_U \mathrm{Map}(F(U), h_{Y/X}(U)) \to \prod_{U\subseteq V} \mathrm{Map}(F(V), h_{Y/X}(U))$ are induced by

$$\begin{array}{ccccc} F(U)\times U & \xleftarrow{\mathrm{incl}^*\times\mathrm{id}} & F(V)\times U & \xrightarrow{\mathrm{id}\times\mathrm{incl}} & F(V)\times V\\ \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$}\\ (s|_U, x) & \longmapsfrom & (s,x) & \longmapsto & (s,x)\end{array}$$

so they are given by

$$\begin{array}{ccccc} \mathrm{Map}(F(U), h_{Y/X}(U)) & \longrightarrow & \mathrm{Map}(F(V), h_{Y/X}(U)) & & \mathrm{Map}(F(V), h_{Y/X})\\ \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$}\\ \alpha_U & \longmapsto & \alpha_U\circ r^F_{UV},\ r^{h_{Y/X}}_{UV}\circ\alpha_V & \longmapsfrom & \alpha_V\end{array}$$

for opens $U\subseteq V\subseteq X$. Hence $f=(\alpha_U)_U\in\prod_U \mathrm{Map}(F(U), h_{Y/X}(U))$ satisfies $fa=fb$ if and only if it is a natural transformation $F\Rightarrow h_{Y/X}$. □

A fancy comment: we can express $\mathrm{Hom}_{\mathbf{PSh}(\mathscr{C})}(F,G)$ as an end

$$\mathrm{Hom}_{\mathbf{PSh}(\mathscr{C})}(F,G) \cong \int_{X\in\mathrm{ob}\,\mathscr{C}} \mathrm{Map}(F(X), G(X)).$$

Remark 3.2.3 · To check that adjoint functors

$$\mathscr{C} \underset{U}{\overset{F}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathscr{D}$$

are inverse equivalences of categories, you need to check that the *unit* and *counit* are isomorphisms. The unit is the natural transformation $\eta : \mathrm{id}_{\mathscr{C}} \Rightarrow UF$ whose component η_A is the transpose of the identity $\mathrm{id}_{FA} : FA \to FA$ under the adjunction $F\dashv U$. Dually, the counit is the natural transformation $\varepsilon : FU \Rightarrow \mathrm{id}_{\mathscr{D}}$ whose component ε_B is the transpose of the identity $\mathrm{id}_{UB} : UB \to UB$ under the adjunction.

Proof (of the restricted equivalence in Theorem 3.2.1). Let $\mathscr{F}$ be a sheaf on X. The unit $\eta_{\mathscr{F}} : \mathscr{F} \Rightarrow h_{\mathrm{sp}(\mathscr{F})/X}$ takes a section $s\in\mathscr{F}(U)$ to the section to $\mathrm{sp}(\mathscr{F})\to X$ given by

$$\begin{tikzcd} \coprod_V \mathscr{F}(V)\times V \arrow[r, two heads, "q"] & \mathrm{sp}(\mathscr{F}) \arrow[r, "p"] & X \\ & U \arrow[u, "(\eta_{\mathscr{F}})_U(s)"] \arrow[ul, bend left] & \end{tikzcd}$$

where the bottom map takes $x\in U$ to $(s,x)_U$. We need to show that the map

$$(\eta_{\mathscr{F}})_U : \mathscr{F}(U) \to \mathbf{Top}_{/X}(U, \mathrm{sp}(\mathscr{F}))$$

is a bijection.

For injectivity, suppose $(\eta_{\mathcal{F}})_U(s) = (\eta_{\mathcal{F}})_U(t)$ for $s, t \in F(U)$. Then $(s, x)_U \approx (t, x)_U$ for all $x \in U$, so there exists a neighbourhood $W_x \subseteq U$ of x on which s and t agree, that is, $s|_{W_x} = t|_{W_x}$. The family $(W_x)_{x \in U}$ covers U, so gluing gives $s = t$.

For surjectivity, suppose $f : U \to \mathrm{sp}(\mathcal{F})$ is a section. For any $x \in U$, pick a lift $(s_x, x)_{V_x} \in \coprod_V F(V) \times V$ of $f(x) \in \mathrm{sp}(\mathcal{F})$ along q, and set $W_x := (\{s_x\} \times V_x)_{V_x}$. Without loss of generality, we may assume V_x is contained in U (otherwise, intersect V_x with U). We showed in Lemma 3.1.5 that the restricted maps

$$W_x \xrightarrow[\cong]{q} q(X) \xrightarrow[\cong]{p} V_x$$

are homeomorphisms. The opens $U_x := f^{-1}(q(W_x))$ cover U as x runs through the points of U and f lifts to a map

$$f|_{U_x} : U_x \to F(V_x) \times V_x, \quad (s_x, x)_{V_x}$$

of the form $(\eta_{\mathcal{F}})_{U_x}(s_x)$. By injectivity, we get $s_x|_{U_x \cap U_y} = s_y|_{U_x \cap U_y}$ for all x and y, so they glue to a section $s \in \mathcal{F}(U)$ such that $f = (\eta_{\mathcal{F}})_U(s)$.

We conclude that the unit is an isomorphism for sheaves. We finish the proof next week by showing that the counit is an isomorphism for local homeomorphisms. $\square$

LECTURE 4

Sheafification, monodromy

4.1 Sheaf/space adjunction (continued)

'Computers don't have noses.'

We still have to prove that the counit of the sheaf/space adjunction is an isomorphism for local homeomorphisms to prove the equivalence $\mathbf{Sh}(X) \simeq \mathbf{LocalHomeo}_{/X}$.

Proof (of the restricted equivalence in Theorem 3.2.1, continued). If $f : Y \to X$ is a local homeomorphism, we have to show that the counit

$$\varepsilon_{Y/X} : \operatorname{sp}(h_{Y/X}) \to Y$$

is an isomorphism in $\mathbf{LocalHomeo}_{/X}$ (or $\mathbf{Top}_{/X}$). Concretely, this map is given by

$$\begin{tikzcd} \coprod_{U \subseteq X} \mathbf{Top}_{/X}(U, Y) \times U \arrow[rr] \arrow[dr, two heads] & & Y \\ & \operatorname{sp}(h_{Y/X}) \arrow[ur, "\varepsilon_{Y/X}"'] & \end{tikzcd}$$

where the top map sends $(s, x)_U$ to $s(x) \in Y$. This map indeed descends to the étalé space $\operatorname{sp}(h_{Y/X})$: if $x \in U \subseteq V$ and $s : V \to Y$ is a section, then $(s, x)_V \sim (s|_U, x)_U$, and both are sent to $s(x)$.

By Lemmas 3.1.3 and 3.1.5, the map $\operatorname{sp}(h_{Y/X}) \to Y$ is a local homeomorphism, so in particular an open map, so it suffices to check that it is bijective.

For injectivity, assume $\varepsilon_{Y/X}([s, x]_U) = \varepsilon_Y([t, y]_V)$. Then $x = y$ since everything is over X, and thus $s(x) = t(x)$ by assumption. Let S be an open neighbourhood of $s(x) = t(x)$ in Y such that f restricts to a homeomorphism $f|_S : S \to f(S)$, and set $W := s^{-1}(S) \cap t^{-1}(S)$. Observe that W contains x. Then $s|_W$ and $t|_W$ are both sections to the injective map $f|_S : S \hookrightarrow X$, so they agree. Hence $(s, x)_U \approx (s|_W, x)_W = (t|_W, x)_W \approx (t, x)_V$ and we conclude that $\varepsilon_{Y/X}$ is injective.

For surjectivity, let $y \in V$ and set $x = f(y) \in X$. Choose an open neighbourhood $V \subseteq Y$ of y such that f restricts to a homeomorphism $f|_V : V \to f(V) =: U$ onto an open subset. Write $s : U \to V$ for its inverse $f|_V^{-1}$; then $s : U \to V \subseteq Y$ is a section with $s(x) = y$, so $\varepsilon_{Y/X}([s, x]_U) = y$, showing that $\varepsilon_{Y/X}$ is surjective.

We conclude that the counit of the sheaf/space adjunction is an isomorphism for local homeomorphisms, finishing the proof of the equivalence of categories $\mathbf{Sh}(X) \simeq \mathbf{LocalHomeo}_{/X}$ of Theorem 3.2.1. □

Definition 4.1.1 · For a space X, the composite functor

$$(-)^\sharp := h_{-/X} \circ \operatorname{sp} : \mathbf{PSh}(X) \to \mathbf{Sh}(X)$$

is called *sheafification*. It comes with a natural map $F \Rightarrow F^\sharp$ for presheaves F on X given by the unit of the adjunction $\operatorname{sp} \dashv h_{-/X}$.

The following result can be seen as the universal property of sheafification.

Corollary 4.1.2 · *Sheafification is left adjoint to the inclusion* $\mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X)$. *That is, if* F *is a presheaf on* X *and* $\mathcal{G}$ *is a sheaf on* X*, then every map* $F \Rightarrow \mathcal{G}$ *of presheaves factors uniquely through the natural map* $F \Rightarrow F^\sharp$.

Proof. Compose the adjunction $\mathrm{sp} \dashv h_{-/X}$ and the equivalence $\mathbf{LocalHomeo}_{/X} \simeq \mathbf{Sh}(X)$. □

The unit of the adjunction is the map $F \Rightarrow F^\sharp$; it is an isomorphism if and only if F is a sheaf.

Example 4.1.3 · For the one-point space $*$, we have $\mathrm{Open}(*) = \{\varnothing \hookrightarrow *\}$, so

$$\mathbf{PSh}(X) = \mathbf{Fun}(\to, \mathbf{Set})$$

is the arrow category of **Set**. As we have seen in the homework, a presheaf F on the one-point space – that is, a map $F(*) \to F(\varnothing)$ – is a sheaf if and only if $F(\varnothing) = *$. Hence we have an equivalence $\mathbf{Sh}(X) \simeq \mathbf{Set}$. Of course, we also have $\mathbf{LocalHomeo}_{/*} \simeq \mathbf{Set}$ since having a local homeomorphism $X \to *$ implies X is discrete.

Example 4.1.4 · Let $S = \{0, 1\}$ be the Sierpiński space with

$$\mathrm{Open}(S) = \{\varnothing \hookrightarrow \{1\} \hookrightarrow S\}.$$

Using a similar argument as in the previous example, we see

$$\mathbf{PSh}(S) \simeq \mathbf{Fun}(\to\to, \mathbf{Set})$$

and

$$\mathbf{Sh}(S) \simeq \mathbf{Fun}(\to, \mathbf{Set}).$$

Exercise 4.1.5 · Check that sheafification on S is given by sending a presheaf F – that is, a pair of maps $F(S) \to F(\{1\}) \to F(\varnothing)$ – to the sheaf given by the map $F(S) \to F(\{1\})$.

4.2 Restriction of sheaves

'You can do this for different values of four, here we do it for five.'

Definition 4.2.1 · Let $U \subseteq X$ be open and let F be a presheaf on X. Then the *restriction* $F|_U$ of F to U is the restriction of the functor F along the inclusion $\mathrm{Open}(U) \hookrightarrow \mathrm{Open}(X)$.

Lemma 4.2.2 · *If* $\mathcal{F}$ *is a sheaf on* X*, then so is its restriction* $\mathcal{F}|_U$ *to an open* $U \subseteq X$*. Under the equivalence* $\mathbf{Sh}(X) \simeq \mathbf{LocalHomeo}_{/X}$*, restriction to* U *corresponds to sending a local homeomorphism* $f : Y \to X$ *to the pullback* $f^{-1}(U) \to U$.

In $\mathbf{LocalHomeo}_{/X}$, we have the subcategory $\mathbf{Cov}_{/X}$ of *covering spaces*: continuous maps $f : Y \to X$ such that X has an open cover $X = \bigcup_{i \in I} U_i$ and there are sets S_i such that $f^{-1}(U_i) \to U_i$ is isomorphic to $U_i \times S_I \to U_i$ (where S_i has the discrete topology) over U_i for all $i \in I$.

Pancake example

Remark 4.2.3 · We will not assume that the space Y in a covering space $Y \to X$ is (path) connected; this condition does appear in the literature with some authors.

Definition 4.2.4 · A sheaf $\mathcal{F}$ on a space X is *locally constant* if there is an open cover $X = \bigcup_{i \in I} U_i$ such that the restriction $\mathcal{F}|_{U_i}$ is a constant sheaf h_{S_i} for some set S_i for all $i \in I$.

We write $\mathbf{Sh}^{\mathrm{lc}}(X)$ for the full subcategory of $\mathbf{Sh}(X)$ on the locally constant sheaves.

Lemma 4.2.5 · *Under the equivalence* $\mathbf{Sh}(X) \simeq \mathbf{LocalHomeo}_{/X}$*, the locally constant sheaves correspond to covering spaces. In other words, this equivalence restricts to an equivalence*

$$\mathbf{Sh}^{\mathrm{lc}}(X) \simeq \mathbf{Cov}_{/X}.$$

The real content is the constant case: we have $\mathcal{F} \cong h_S = h_{X \times X/X}$ if and only if $\mathrm{sp}(\mathcal{F}) \cong X \times S$ over X.

Four-to-one cover of the circle example

4.3 Review of monodromy

'If you've ever been in a parking garage, you know what I mean.'

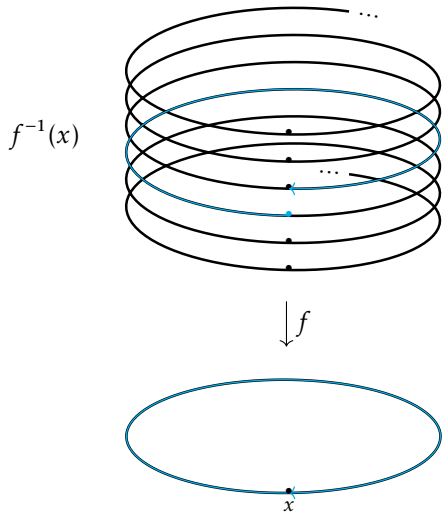

Figure 4.1 · Unique path lifting in the cover of the circle by the helix

Definition 4.3.1 · Let X be a topological space. Then X is

(i) *locally path connected* if every open neighbourhood $x \in U$ of any $x \in X$ contains a path connected open neighbourhood $x \in V \in U$.

(ii) *semi-locally path connected* if for every $x \in X$, there exists an open neighbourhood $x \in U$ such that the map $\pi_1(U, x) \to \pi_1(X, x)$ is trivial.

Definition 4.3.2 · Let X be a topological space and $x \in X$. The *fibre functor* is defined by

$$\begin{aligned} F_x : \mathbf{Cov}_{/X} &\to \pi_1(X, x)\mathbf{Set} \\ (Y \xrightarrow{f} X) &\mapsto f^{-1}(x), \end{aligned}$$

where a loop $\gamma \in \pi_1(X, x)$ acts on $f^{-1}(x)$ by unique path lifting.

Theorem 4.3.3 (monodromy correspondence) · *Let X be a topological space and $x \in X$.*

(i) If X is path connected and locally path connected, then

$$F_x : \mathbf{Cov}_{/X} \to \pi_1(X, x)\mathbf{Set}$$

is fully faithful.

(ii) If X is moreover semi-locally simply connected, then F_x is an equivalence of categories.

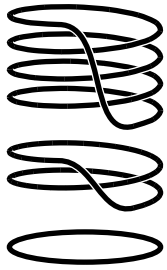

Figure 4.2 · Covering of the circle by a disjoint union of the four-to-one cover and the two-to-one cover.

Proof (outline).

Maybe type up the details Marcel

- Any covering $f : Y \to X$ is locally path connected (easy). It is furthermore a disjoint union $\coprod_{i \in I} Y_i$ of path connected spaces Y_i [Munoo, Theorem 25.4].
- Any map $Y \xrightarrow{\varphi} Y' \in \mathbf{Cov}_{/X}$ is itself a covering [Munoo, Lemma 80.2(a)].
- F_x is faithful: if two maps

$$\begin{tikzcd} Y \arrow[rr, shift left, "\varphi"] \arrow[rr, shift right, "\varphi'"'] \arrow[dr, "f"'] & & Y \arrow[dl, "f'"] \\ & X & \end{tikzcd}$$

  agree on $f^{-1}(x)$ then they agree: for $y \in Y$ arbitrary, choose a path γ starting at $f(y)$ and ending at x. By unique path lifting, this gives a path $\overline{\gamma}$ from y to y' for some $y' \in f^{-1}(x)$. Then $\varphi(\overline{\gamma})$ and $\varphi'(\overline{\gamma})$ are both lifts of γ to paths ending at $\varphi'(y') = \varphi(y')$ so by uniqueness they are the same path and their starting points are the same, i.e. $\varphi(y) = \varphi'(y)$.
- That F_x is full follows from the lifting lemma [Munoo, Lemma 79.1].
- F_x is essentially surjective if X is semi-locally simply connected: every $S \in \pi_1(X, x)\mathbf{Set}$ is $S \cong \cup_{i \in I} S_i$ where $\pi_1(X, x)$ acts transitively on S_i. Then

explain

$$S_i \cong \pi_1(X, x)/H_i$$

  for $H_i = \mathrm{Stab}(s_i)$ for any $s_i \in S_i$. There exists a covering $Y_i \xrightarrow{f_i} X$ with $\pi_1(Y_i, y_i) \hookrightarrow \pi_1(X, x)$, one subgroup H_i [Munoo, Theorem 82.1], so $S_i \cong F_x(Y_i \xrightarrow{f_i} X)$ and $S = F_x(\coprod_{i \in I} Y_i \to X)$. □

Example 4.3.4 · Let $X = S^1$. Let Y_1 be the four-to-one covering of S^1, and let Y_2 be the double covering of S^1. Let $Y = Y_1 \coprod Y_2$. Then $Y \to X$ corresponds to the set $\{1, 2, 3, 4, 5, 6\}$ where the single loop around the circle $1 \in \pi_1(S^1, x)$ acts by $(12)(3456)$.

Diagram below should have labels etc

The following diagram summarises the situation if X is a 'nice' space:

$$\begin{tikzcd} \mathbf{Sh}(X) \arrow[r, "\simeq"] & \mathbf{LocalHomeo}_{/X} & \\ \mathbf{Sh}^{\mathrm{lc}}(X) \arrow[u, hook] \arrow[r, "\simeq"'] & \mathbf{Cov}_{/X} \arrow[u, hook] \arrow[r, "\simeq"'] & \pi_1(X, x)\mathbf{Set} \end{tikzcd}$$

This diagram raises the question: what should there be in the top right spot? Towards the end of the course, we will give a partial answer which goes by the name of *exodromy*.

LECTURE 5

Pullback and pushforward

'There is a risk you might learn something – beware!'

The goal of this lecture is to define a pair of adjoint functors

$$\mathbf{Sh}(Y) \underset{f_*}{\overset{f^*}{\underset{\longrightarrow}{\overset{\longleftarrow}{\perp}}}} \mathbf{Sh}(X)$$

for a continuous map $f : Y \to X$, called *pullback* and *pushforward*. The strategy will be to first define these operations for presheaves, and then restrict to sheaves. One of the functors will already send sheaves to sheaves, for the other we will postcompose the functor on the level of presheaves with sheafification.

5.1 Pushforward

A continuous map $f : Y \to X$ induces a functor $f^{-1} : \mathrm{Open}(X) \to \mathrm{Open}(Y)$ which sends an open set $U \subseteq X$ to the open set $f^{-1}(U) \subseteq Y$. If $U \subseteq V \subseteq X$ are open sets, then $f^{-1}(U) \subseteq f^{-1}(V) \subseteq Y$, so this is indeed functorial.

Definition 5.1.1 · The *pushforward* of a presheaf G on Y along f is the composite presheaf

$$f_*G : \mathrm{Open}(X)^{\mathrm{op}} \xrightarrow{f^{-1}} \mathrm{Open}(Y)^{\mathrm{op}} \xrightarrow{G} \mathbf{Set}$$

on X. Explicitly, the value of the pushforward f_*G on an open $U \subseteq X$ is $(f_*G)(U) = G(f^{-1}(U))$.

The pushforward functor $f_* : \mathbf{PSh}(Y) \to \mathbf{PSh}(X)$ is given by precomposition by the functor $f^{-1} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Open}(Y)^{\mathrm{op}}$, and so is indeed seen to be functorial.

Lemma 5.1.2 · *If $\mathcal{G}$ is a sheaf on Y, then the pushforward $f_*\mathcal{G}$ of $\mathcal{G}$ along f is a sheaf on X. In particular, pushforward restricts to a functor $f_* : \mathbf{Sh}(Y) \to \mathbf{Sh}(X)$ on the level of sheaves.*

Proof. Let $U = \bigcup_{i \in I} U_i$ be an open cover in X. Then $f^{-1}(U) = \bigcup_{i \in I} f^{-1}(U_i)$ is an open cover in Y. Applied to this cover, the sheaf condition for $\mathcal{G}$ gives us an equaliser diagram

$$\mathcal{G}(f^{-1}(U)) \longrightarrow \prod_{i \in I} \mathcal{G}(f^{-1}(U_i)) \rightrightarrows \prod_{i,j \in I} \mathcal{G}(f^{-1}(U_i \cap U_j))$$

(where we have rewritten $f^{-1}(U_i) \cap f^{-1}(U_j) = f^{-1}(U_i \cap U_j)$), giving the sheaf condition for the pushforward $f_*\mathcal{G}$. □

5.2 Pullback

'Just sheafify the hell out of everything.'

The *pullback* of a sheaf on X along f should be a sheaf on Y. One way to approach the problem of constructing a presheaf on Y from a presheaf $\mathcal{F}$ on X, would be to try extending it along f^{-1} (and this is what we will attempt):

$$\begin{tikzcd} \mathrm{Open}(X)^{\mathrm{op}} \arrow[rr, "\mathcal{F}"] \arrow[dr, "f^{-1}"'] & & \mathbf{Set} \\ & \mathrm{Open}(Y)^{\mathrm{op}} \arrow[ur, dashed, "?"'] & \end{tikzcd}$$

In general, there might not be such an on-the-nose extension, but we can approximate an extension by considering extensions *up to* a natural transformation. There could be many such approximations, so we want to consider the 'best possible approximation' in some suitable sense.

In category theory, the general problem of approximating an extension of a functor along another functor is studied using *Kan extensions*; we refer to [Rie16, Chapter 6] for an introduction of Kan extensions, whose definition we actually do not need to know for our current purposes. Here we will apply the general theory to our case to define the pullback of sheaves.[1]

Proposition 5.2.1 · *Left Kan extensions of presheaves on X (functors $\mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Set}$) along the functor $f^{-1} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathrm{Open}(Y)^{\mathrm{op}}$ exist, and assemble into a functor*

$$\mathrm{Lan}_{f^{-1}} : \mathbf{PSh}(X) \to \mathbf{PSh}(Y)$$

which is left adjoint to the pushforward functor f_*:

$$\begin{tikzcd} \mathbf{PSh}(Y) \arrow[r, shift right, "f_*"'] \arrow[r, phantom, "\perp"] & \mathbf{PSh}(X) \arrow[l, shift right, "\mathrm{Lan}_{f^{-1}}"'] \end{tikzcd}$$

Moreover, the left Kan extension along f^{-1} of a presheaf F on X is given on opens $U \subseteq Y$ by

$$\mathrm{Lan}_{f^{-1}} F(U) = \operatorname*{colim}_{f^{-1}(W) \supseteq U} F(W) \tag{5.1}$$

where W ranges over opens of X (more precisely, the colimit diagram is indexed by the full subcategory of $\mathrm{Open}(X)$ on those W such that $f^{-1}(W) \supseteq U$, or equivalently $W \supseteq f(U)$).

Proof. This is a special case of [Rie16, Corollary 6.2.6]; the only nontrivial step in applying the general result to this special case is recognising that the comma category $f^{-1} \downarrow U$ for $U \in \mathrm{Open}(Y)^{\mathrm{op}}$ is the described index category, but this verification is elementary enough to be left to the reader. □

We would also like a description of what $\mathrm{Lan}_{f^{-1}}$ does on maps. By a computation of the colimit in (5.1), an element of $\mathrm{Lan}_{f^{-1}} F(U)$ is given by $[s]_W$ for some section $s \in F(W)$ for some $W \supseteq f(U)$, where $[s]_W = [s']_{W'}$ if and only if there exists an open $W'' \subseteq W \cap W'$ containing $f(U)$ such that $s|_{W''} = s'|_{W''}$. Unravelling the construction in [Rie16, Theorem 6.2.1], we see that the map induced by opens $U \subseteq V$ in Y is given by

$$\operatorname*{colim}_{f^{-1}(W) \supseteq V} F(W) \to \operatorname*{colim}_{f^{-1}(W) \supseteq U} F(W), \quad [s]_W \mapsto [s]_W.$$

We may now define the pullback as follows:

[1] One can also construct and prove everything by hand, as was done in class, but this is rather tedious. We wish to illustrate with the following exposition that all the arguments will be completely formal.

Definition 5.2.2 · The *pullback* $f^{\circledast}F$ of a presheaf F on X along f is the left Kan extension of F along f^{-1}. The pullback functor is denoted $f^{\circledast} := \mathrm{Lan}_{f^{-1}} : \mathbf{PSh}(X) \to \mathbf{PSh}(Y)$.

We have defined the pullback functor in such a way that it is left adjoint to the pushforward of presheaves.

We use the circled asterisk $\circledast$ in the notation for the pullback of presheaves to distinguish it from the pullback of sheaves, which we will define momentarily, and for which we use a normal asterisk.[2] Unlike the pushforward, namely, the pullback of presheaves does not directly restrict to sheaves; that is to say, there are sheaves which are sent by $f^{\circledast}$ to a presheaf which does not satisfy the sheaf condition, as witnessed by the following counterexample:

Example 5.2.3 · Let Y be the two-point space with the discrete topology, and let X be the one-point space, with the unique map $f : Y \to X$. We can easily put a sheaf on X, for example by defining $\mathcal{F}(\varnothing) = *$ and $\mathcal{F}(X) = \{42, 43, 44\}$. The pullback in presheaves at any point $\{*\} \subseteq Y$ is $f^{\circledast}(\{*\}) = \mathrm{colim}_{f^{-1}(W) \supseteq *} \mathcal{F}(W) = \mathcal{F}(X) = \{42, 43, 44\}$. But the pullback in presheaves at Y itself is also $\mathcal{F}(X) = \{42, 43, 44\}$. Check that the sheaf condition doesn't hold for the cover of Y given by two opens, one containing precisely each point: the gluing condition on $\{42, 43, 44\} \times \{42, 43, 44\}$ is void because the points do not intersect.

Definition 5.2.4 · The *pullback* $f^*\mathcal{F}$ of a sheaf $\mathcal{F}$ on X along f is the sheaf

$$f^*\mathcal{F} := (f^{\circledast}\mathcal{F})^{\sharp},$$

the sheafification of the presheaf pullback of $\mathcal{F}$. The pullback functor of sheaves is thus the composite

$$f^* : \mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X) \xrightarrow{f^{\circledast}} \mathbf{PSh}(Y) \xrightarrow{(-)^{\sharp}} \mathbf{Sh}(Y).$$

Lemma 5.2.5 · *Pushforward and pullback of sheaves along f define an adjunction*

$$\begin{tikzcd} \mathbf{Sh}(Y) \arrow[r, shift right, "f_*"'] \arrow[r, phantom, "\perp"] & \mathbf{Sh}(X) \arrow[l, shift right, "f^*"'] \end{tikzcd}$$

Proof. Compose the adjunctions

$$\begin{tikzcd} \mathbf{Sh}(Y) \arrow[r, shift right, "f_*"'] \arrow[r, phantom, "\perp"] \arrow[rr, bend right, "f_*"'] & \mathbf{PSh}(Y) \arrow[l, shift right, "(-)^{\sharp}"'] \arrow[r, shift right, "f_*"'] \arrow[r, phantom, "\perp"] & \mathbf{PSh}(X) \arrow[l, shift right, "f^{\circledast}"'] \arrow[l, bend right, "f^*"'] \\ & & \mathbf{Sh}(X) \arrow[u, hook] \end{tikzcd}$$

of Corollary 4.1.2 and Proposition 5.2.1. □

Between the sets $\mathrm{Hom}_{\mathbf{Sh}(Y)}(f^*\mathcal{F}, \mathcal{G})$ and $\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathcal{F}, f_*\mathcal{G})$ which are naturally isomorphic by the adjunction, there is also an 'intermediate' set of maps, called *f-maps*; see [Sta24, Lemma 008K].

5.3 Stalks, germs, skyscrapers

Definition 5.3.1 · Let $i_x : \{x\} \hookrightarrow X$ be the inclusion of a point in a space. Then $i_x^*\mathcal{F}$ is the *stalk* $\mathcal{F}_x$ of $\mathcal{F}$ at x; this is a sheaf on a point, so equivalently a set by Example 4.1.3 (the value of the sheaf on the whole space). Unravelling definitions, we have

$$\mathcal{F}_x = \operatorname*{colim}_{U \ni x} \mathcal{F}(U).$$

[2] In class, we used the notation $f^{\mathrm{pre},*}$ for $f^{\circledast}$, which we do not like.

An element of the stalk at x is represented by a *germ* $[s]_U$ for $U \ni x$ and $s \in \mathcal{F}(U)$, where $[s]_U = [t]_V$ if and only if there exists an open neighbourhood $W \subseteq U \cap V$ of x such that $s|_W = t|_W$.

Note that the stalk $\mathcal{F}_x$ is the fibre of $\mathrm{sp}(X) \to X$ over x.

Example 5.3.2 (orientation sheaf of a manifold) · Let M be an n-dimensional topological manifold, by which we mean a Hausdorff space that is locally homeomorphic to $\mathbb{R}^n$ (i.e., every point of M admits a neighbourhood homeomorphic to $\mathbb{R}^n$). For a subset $K \subseteq M$, we write

$$H_k(M \mid K; R) := H_k(M, M \setminus K; R)$$

for the kth homology of M relative to $M \setminus K$ with coefficients in a commutative ring R.

A *local R-orientation* at a point $x \in M$ is a generator μ_x of $H_n(M \mid x; R) \cong R$ (note that we take homology in degree n, the dimension of the manifold M). That $H_n(M \mid x; R)$ is isomorphic to R (as R-modules) follows from excision; choosing a generator corresponds exactly to choosing such an isomorphism. An *R-orientation* of M is a family of local orientations $(\mu_x)_{x \in M}$ with the property that every point $y \in M$ admits a compact neighbourhood K and an element $\mu_K \in H_n(M \mid K; R)$ such that μ_K restricts to μ_x for any $x \in K$ under the map $H_n(M \mid K; R) \to H_n(M \mid x; R)$ induced by the inclusion $M \setminus K \to M \setminus \{x\}$.

There is a sheaf $\mathcal{O}_M$ of R-modules, the *orientation sheaf* of M, defined by

$$\mathcal{O}_M(U) := H_n(M \mid U; R)$$

for $U \subseteq M$ open. The restriction map for opens $U \subseteq V \subseteq M$ is induced by the inclusion of pairs $(M, M \setminus V) \hookrightarrow (M, M \setminus U)$. The stalk of the orientation sheaf $\mathcal{O}_M$ at $x \in M$ is the R-module $H_n(M \mid x; R)$.

Remark 5.3.3 · One can show (as a generalisation of an exercise in Homework 3) that the étalé space of the pullback can be obtained as a fibre product. More precisely, given $f : Y \to X$, the diagram

$$\begin{tikzcd} \mathrm{sp}(f^*\mathcal{F}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{sp}(\mathcal{F}) \arrow[d] \\ Y \arrow[r] & X \end{tikzcd}$$

is a pullback square. This is used in to define f^*.

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Given a map $\alpha : \mathcal{F} \to \mathcal{G}$ between sheaves on a space X and a point $x \in X$, there is an induced map on stalks $\alpha_x : \mathcal{F}_x \to \mathcal{G}_x$, because $\mathcal{G}_x$ is a cocone over

$$\{\mathcal{F}(U) \mid U \in \mathrm{Open}(X), x \in U\},$$

of which $\mathcal{F}_x$ is a colimit.

Lemma 5.3.4 · *Let $\alpha : \mathcal{F} \to \mathcal{G}$ be a map in* $\mathbf{Sh}(X)$. *Then α is an isomorphism of sheaves if and only if $\alpha_x : \mathcal{F}_x \to \mathcal{G}_x$ is a bijection for all $x \in X$.*

Proof. The map $\mathrm{sp}(\alpha) : \mathrm{sp}(\mathcal{F}) \to \mathrm{sp}(\mathcal{G})$ is a local homeomorphism over X, because the maps $\mathrm{sp}(\mathcal{F}) \to X$ and $\mathrm{sp}(\mathcal{G}) \to X$ are local homeomorphisms. In particular, $\mathrm{sp}(\alpha)$ is an open map. So it is invertible if and only if it is a bijection. Check that this is equivalent to requiring that the map on the fibres over x is a bijection for all $x \in X$. Since the stalks $\mathcal{F}_x$ and $\mathcal{G}_x$ are exactly these fibres, we are done. □

Definition 5.3.5 · Let $\{x\} \to X$ be the inclusion of a point $x \in X$. Let $S \in \mathbf{Set}$. The *skyscraper sheaf* at x with value S is $i_{x,*}\underline{S}$, the pullback of the constant sheaf. We denote it $i_{*,S}$.

Example 5.3.6 · Let $\{0\} \to \mathbb{R}$ be the inclusion of zero into the real numbers. Let $S = \{a, b\}$. Then the skyscraper sheaf is

$$i_{*,S}(U) = S(U \cap 0) = \begin{cases} \{a, b\} & \text{if } 0 \in U \\ * & \text{if } x \notin U. \end{cases}$$

In Homework 2, we saw that the étalé space of this sheaf is the line with two origins.

LECTURE 6

Sheaves of abelian groups

6.1 Sheaves of abelian groups

'Why is he erasing an empty board,
you might ask.'

Almost everything we have done so far works for sheaves with algebraic structure, for example:
- Sheaves of abelian groups ($\rightsquigarrow$ homological algebra)
- Sheaves of commutative rings (algebraic geometry)
- Sheaves of commutative monoids (useful in logarithmic geometry (?))

You can also do more general things:
- Sheaves of topological rings
- Sheaves of Banach algebras
- Sheaves of categories (*stacks in categories*)
- Sheaves of groupoids (*stacks*)
- Sheaves of homotopy types/spaces/anima/∞-groupoids (∞-*stacks*)

We should warn that the sheaf condition should be modified for higher-categorical sheaves.

Definition 6.1.1 · Let $\mathscr{C}$ be a category. A *presheaf on* $X \in \mathbf{Top}$ *with values in* $\mathscr{C}$ is a functor $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathscr{C}$. If $\mathscr{C}$ is complete (or at least the products of the sheaf condition), then a sheaf on X with values in $\mathscr{C}$ is a presheaf such that the sheaf condition holds in $\mathscr{C}$.

We denote the categories of such objects $\mathbf{PSh}(X, \mathscr{C})$ and $\mathbf{Sh}(X, \mathscr{C})$ respectively. In the case $\mathscr{C}$ is $\mathbf{Ab}$, we may use the notation $\mathbf{PAb}(X) := \mathbf{PSh}(X, \mathbf{Ab})$ and $\mathbf{Ab}(X) := \mathbf{Sh}(X, \mathbf{Ab})$.

We will need a description of limits in $\mathbf{Ab}$.

Lemma 6.1.2 ([MM94, Section v.1]) · *The forgetful functor* $\mathbf{Ab} \to \mathbf{Set}$ *creates limits.*

Example 6.1.3 · The product of a family $(A_i)_{i\in I}$ of abelian groups is given by the set $\prod_{i\in I} A_i$ with coordinatewise addition. Equalisers in $\mathbf{Ab}$ (and in fact in any additive category with all kernels) are given by

$$\mathrm{eq}(f, g) = \{ a \in A \mid f(a) = g(a) \} = \ker(f - g).$$

However, colimits are not the same as in $\mathbf{Set}$. The coproduct of a family $(A_i)_{i\in I}$ of abelian groups is the direct sum

$$\bigoplus_{i\in I} A_i = \{ (a_i)_{i\in I} \in \prod_{i\in I} A_i \mid \#\{ i \in I \mid a_i \neq 0 \} < \infty \},$$

and coequalisers in $\mathbf{Ab}$ are given by $\mathrm{coeq}(f, g) = \mathrm{coker}(f - g)$.

Corollary 6.1.4 · *A presheaf* $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Ab}$ *is a sheaf if and only if the composite*

$$\mathrm{Open}(X)^{\mathrm{op}} \xrightarrow{F} \mathbf{Ab} \xrightarrow{U} \mathbf{Set}$$

is a sheaf, where $U : \mathbf{Ab} \to \mathbf{Set}$ *is the forgetful functor.*

Definition 6.1.5 · A small category $\mathcal{I}$ is *filtered* if $\mathcal{I} \neq \varnothing$ and the following conditions hold:

(i) For all $i, j \in \mathcal{I}$ there exists a $k \in \mathcal{I}$ and arrows $i \to k \leftarrow j$

(ii) For $u, v : i \to j$ in $\mathcal{I}$ there exists a $k \in \mathcal{I}$ and $w : j \to k$ such that $wu = wv$.

Dually, there is a notion of *cofiltered category*.

Example 6.1.6 · A poset P always satisfies (ii), so it is filtered if and only if it is nonempty and every pair of elements has an upper bound. If I is a set, then the poset $\mathcal{P}_{\text{fin}}(I) := \{ J \subseteq I \mid I \text{ finite} \}$ of finite subsets of I ordered by inclusion is filtered, since $\mathcal{P}_{\text{fin}}(I)$ contains the empty set and two finite subsets $J, J' \subseteq I$ have an upper bound $J \cup J'$.

Exercise 6.1.7 · Show that a category $\mathcal{I}$ is filtered if and only if every finite diagram $D : \mathcal{J} \to \mathcal{I}$ in $\mathcal{I}$ has a cocone.

Definition 6.1.8 (Only the notation was introduced in the lecture) · A *filtered colimit*, denoted $\varinjlim$, is the colimit of a functor $F : \mathcal{I} \to \mathcal{C}$ where $\mathcal{I}$ is a filtered category. A *cofiltered limit*, denoted $\varprojlim$, is the limit of a functor $F : \mathcal{I} \to \mathcal{C}$ where $\mathcal{I}$ is a cofiltered category.

Lemma 6.1.9 ([MM94, Section IX.1]) · *The forgetful functor* $U : \mathbf{Ab} \to \mathbf{Set}$ *creates filtered colimits.*

Example 6.1.10 · The following are examples of filtered limits and colimits, in some cases the indexing category is omitted in the notation.

- We saw that
$$\bigoplus_{i \in I} A_i \cong \varinjlim_{J \in \mathcal{P}_{\text{fin}}(I)} \oplus_{j \in J} A_j.$$
- The filtered colimit of the rings $\frac{1}{n}\mathbb{Z}$ where the diagram is indexed by the divisibility poset is
$$\varinjlim_n \frac{1}{n}\mathbb{Z} = \mathbb{Q}.$$
- The filtered colimit of the field extensions of $\mathbb{Q}$ is
$$\varinjlim_{\mathbb{Q} \to K \text{ finite}} K = \overline{\mathbb{Q}}.$$
- The filtered limit of the rings $\mathbb{Z}/p^n\mathbb{Z}$ for p a prime is the ring
$$\varprojlim_n \mathbb{Z}/p^n\mathbb{Z} = \mathbb{Z}_p$$
of the p-adic integers.
- The stalk of a sheaf $\mathcal{F}$ at a point x is a filtered colimit:
$$\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U).$$
Indeed the indexing category $\{ U \in \operatorname{Open}(X)^{\text{op}} \mid x \in U \}$ is filtered: if $x \in U, V \subseteq X$ are open then $x \in U \cap V$.
- Likewise, the pullback of a presheaf $\mathcal{F}$ is given by
$$f^{\circledast}\mathcal{F}(V) = \varinjlim_{f(V) \subseteq U} \mathcal{F}(U),$$
a filtered colimit.

6.2 Abelian group objects

'I'm only familiar with this for ∞-categories, not for 1-categories.'

Definition 6.2.1 · Let $\mathscr{C}$ be a category with finite products. An *abelian group object* in $\mathscr{C}$ is a quadruple $(A, m, i, 0)$ of an object $A \in \mathscr{C}$ and morphisms

$$m : A \times A \to A, \quad i : A \to A, \quad 0 : * \to A$$

such that the following diagrams commute:

(i) Associativity:

$$\begin{tikzcd} A \times A \times A \arrow[r, "m \times \mathrm{id}"] \arrow[d, "\mathrm{id} \times m"'] & A \times A \arrow[d, "m"] \\ A \times A \arrow[r, "m"'] & A \end{tikzcd}$$

(ii) Commutativity:

$$\begin{tikzcd} A \times A \arrow[rr, "\mathrm{swap}"] \arrow[dr, "m"'] & & A \times A \arrow[dl, "m"] \\ & A & \end{tikzcd}$$

(iii) Identity:

$$\begin{tikzcd} A \times * \arrow[r, "\mathrm{id} \times 0"] \arrow[d, "\mathrm{pr}_1"'] & A \times A \arrow[d, "m"] & * \times A \arrow[l, "0 \times \mathrm{id}"'] \arrow[d, "\mathrm{pr}_2"] \\ A \arrow[r, "\mathrm{id}"'] & A & A \arrow[l, "\mathrm{id}"] \end{tikzcd}$$

(iv) Inverse:

$$\begin{tikzcd} * \arrow[d, "0"'] & & & A \arrow[lll] \arrow[rrr] \arrow[d, "\Delta"] & & & * \arrow[d, "0"] \\ A & A \times A \arrow[l, "m"] & & A \times A \arrow[ll, "i \times \mathrm{id}"] \arrow[r, "\mathrm{id} \times i"'] & A \times A \arrow[r, "m"'] & A \end{tikzcd}$$

Write $\mathbf{Ab}(\mathscr{C})$ for the *category of abelian group objects* in $\mathscr{C}$. A map $f : (A, m, i, 0) \to (B, m', i', 0')$ in $\mathbf{Ab}(\mathscr{C})$ is a map $f : A \to B$ in $\mathscr{C}$ such that $f \circ m = m' \circ (f \times f)$, that is, such that f commutes with the group operation m.

Using the diagrams above, one can show that a map of abelian group objects in $\mathscr{C}$ also commutes with inverses and the unit.

Remark 6.2.2 · The category $\mathbf{Ab}$ of abelian groups is precisely the category $\mathbf{Ab}(\mathbf{Set})$ of abelian group objects in $\mathbf{Set}$.

Lemma 6.2.3 · *We have the following equivalences of categories:*

(i) $\mathbf{Ab}(\mathbf{PSh}(X)) \cong \mathbf{PAb}(X)$

(ii) $\mathbf{Ab}(\mathbf{Sh}(X)) \cong \mathbf{Ab}(X)$

Proof (sketch). (i) A presheaf $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Ab}$ consists of $F(U) \in \mathbf{Ab}$ for all $U \in \mathrm{Open}(X)^{\mathrm{op}}$ with restriction maps $r_{UV} : F(V) \to F(U)$ in $\mathbf{Ab}$. This means that the group operation $F(U) \times F(U) \to F(U)$ is natural, that is, if $U \subseteq V$ then

$$\begin{tikzcd} F(V) \times F(V) \arrow[r, "m_V"] \arrow[d, "r_{UV} \times r_{UV}"'] & F(V) \arrow[d, "r_{UV}"] \\ F(U) \times F(U) \arrow[r, "m_U"'] & F(U) \end{tikzcd}$$

So $m : F \times F \to F$ is natural. Likewise, 0 and i are natural transformations (follow your nose). Conversely, an abelian object in $\mathbf{PSh}(X)$ is a presheaf of abelian groups (why?).

(ii) By Corollary 6.1.4, a presheaf $F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Ab}$ is a sheaf if and only if the composite $U \circ F : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Set}$ is a sheaf, where $U : \mathbf{Ab} \to \mathbf{Set}$ is the forgetful functor. Thus for any sheaf $\mathcal{F} \in \mathbf{Ab}(X)$, $U \circ \mathcal{F}$ is a sheaf of sets which has abelian group object structure by (i). Conversely, $(\mathcal{F}, m, i, 0) \in \mathbf{Ab}(\mathbf{Sh})$ is the data of a sheaf of abelian groups, since it is a presheaf of abelian groups by (i) and thus a sheaf since it is a sheaf on sets. □

Remark 6.2.4 · There is a 'tensor product' $\otimes$ on presentable categories for which we (probably) have $\mathbf{Ab}(X) \simeq \mathbf{Sh}(X) \otimes \mathbf{Ab}$.

Remark 6.2.5 · We can do everything we know for sheaves of abelian groups as well:

- We have an adjunction:

$$\mathbf{PAb}(X) \underset{h_{-/X}}{\overset{\mathrm{sp}}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathbf{Ab}(\mathbf{Top}_{/X})$$

  which restricts to an equivalence:

$$\mathbf{Ab}(X) \underset{h_{-/X}}{\overset{\mathrm{sp}}{\underset{\longleftarrow}{\overset{\longrightarrow}{\simeq}}}} \mathbf{Ab}(\mathbf{LocalHomeo}_{/X})$$

- Monodromy:

$$\mathbf{Ab}(\mathbf{Sh}^{\mathrm{lc}}(X)) \simeq \mathbf{Rep}_{\mathbb{Z}}(\pi_1(X, x)),$$

  where $\mathbf{Rep}_{\mathbb{Z}}(G)$ denotes the category of G-representations in $\mathbf{Mod}_{\mathbb{Z}}$. We may also denote $\mathbf{Ab}(\mathbf{Sh}^{\mathrm{lc}}(X))$ by $\mathbf{Ab}^{\mathrm{lc}}(X) \simeq \mathbf{Mod}_{\mathbb{Z}[\pi_1(X,x)]}$, where $\mathbb{Z}[\pi_1(X, x)]$ is the group algebra of $\pi_1(X, x)$.
- Pushforward and pullback: for $f : Y \to X$ we have adjunctions

$$\mathbf{PAb}(Y) \underset{f_*}{\overset{f^{\circledast}}{\underset{\longrightarrow}{\overset{\longleftarrow}{\perp}}}} \mathbf{PAb}(X)$$

  and

$$\mathbf{Ab}(Y) \underset{f_*}{\overset{f^*}{\underset{\longrightarrow}{\overset{\longleftarrow}{\perp}}}} \mathbf{Ab}(X)$$

- All the functors $f_*, f^{\circledast}, f^*, (-)^\sharp, h_{-/X}$ commute with the forgetful functors (see [Sta24, Lemma 0085] for a hands-on proof that $(-)^\sharp$ does).
- A map $f : \mathcal{F} \to \mathcal{G}$ in $\mathbf{Ab}(X)$ is an isomorphism if and only if the induced map $f_x : \mathcal{F}_x \to \mathcal{G}_x$ on stalks is an isomorphism for all $x \in X$.

In order to do homological algebra for sheaves of abelian groups, we need a notion of *exactness* for sequences of sheaves of abelian groups. Since the image presheaf of map of sheaves is in general not a sheaf itself, we define exactness in $\mathbf{Ab}(X)$ via stalks.

Definition 6.2.6 · A sequence

$$\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$$

of sheaves of abelian groups on a space X is *exact at* $\mathcal{G}$ if the induced sequence

$$\mathcal{F}_x \xrightarrow{f_x} \mathcal{G}_x \xrightarrow{g_x} \mathcal{H}_x$$

of stalks is exact at $\mathcal{G}_x$ for all points $x \in X$.

Remark 6.2.7 · Coincidentally, the reason to define exactness via the stalks – that the image presheaf is not a sheaf – is also the reason the global sections functor (evaluation in the global sections) is generally only left exact, and thus gives rise to a right derived functor. Right deriving the global sections functor on a sheaf $\mathcal{F} \in \mathbf{Ab}(X)$ allows us to compute sheaf cohomology, denoted $H^i(X; \mathcal{F})$.

Example 6.2.8 · Let $0 \to A \to B \to C \to 0$ be a short exact sequence in **Ab**. Then the sequence

$$0 \to \underline{A} \to \underline{B} \to \underline{C} \to 0$$

of constant sheaves is exact. Indeed, $(\underline{Z})_x \cong Z$ for any $x \in X$ and any abelian group or set Z, since this is the fibre of $Z \times X \xrightarrow{\mathrm{pr}_X} X$ at $x \in X$.

Example 6.2.9 · Let $\iota : \{0\} \hookrightarrow \mathbb{R}$, $A \in$ **Ab**. Then $\iota_* A$ has etale space,

draw line with $\#A$ origins

LECTURE 7

Limits and colimits of (pre)sheaves

Definition 7.0.1 · A *diagram* in a category $\mathscr{C}$ is a functor $D : \mathscr{J} \to \mathscr{C}$ from a small category $\mathscr{J}$.

In this lecture, we will show that every diagram $D : \mathscr{J} \to \mathbf{PSh}(X)$ or $D : \mathscr{J} \to \mathbf{PAb}(X)$ has a limit and colimit, and we will describe them explicitly.

7.1 Limits and colimits of sets

We will begin by treating limits and colimits in **Set**. For a more complete treatment of this topic, we refer the reader to [Rie16, Chapter 3.2].

Proposition 7.1.1 · *The limit of $D : \mathscr{J} \to$ **Set** exists and is given by*

$$\lim_{i \in \mathscr{J}} D(i) = \{(a_i)_{i \in \mathscr{J}} \in \prod_{i \in \mathscr{J}} D(i) \mid D(\varphi)(a_i) = a_j \textit{ for all } \varphi : i \to j \textit{ in } \mathscr{J}\}$$

Proof. Denote our candidate limit set S. There are maps $S \xrightarrow{\pi_i} D(i)$ for all $i \in \mathscr{J}$. For any $\varphi : i \to j$ in $\mathscr{J}$ the diagram

$$\begin{tikzcd} & S \arrow[dl, "\pi_i"'] \arrow[dr, "\pi_j"] & \\ D(i) \arrow[rr] & & D(j) \end{tikzcd}$$

commutes. This commutativity is directly by construction of our set S. Given any cone $(C \xrightarrow{\pi_i} D(i))_{i \in \mathscr{J}}$ there exists a unique map of cones to $(S \xrightarrow{\pi_i} D(i))_{i \in \mathscr{J}}$ making the diagram

$$\begin{tikzcd} & C \arrow[d, dashed] \arrow[ddl, bend right] \arrow[ddr, bend left] & \\ & S \arrow[dl, "\pi_i"'] \arrow[dr, "\pi_j"] & \\ D(i) \arrow[rr] & & D(j) \end{tikzcd}$$

commute. It is given by $c \mapsto (\varphi_i(c))_{i \in \mathscr{J}}$.

□

Remark 7.1.2 · The limit of a diagram $D : \mathscr{J} \to$ **Set** is the equalizer of

$$\prod_{i \in \mathscr{J}} D(i) \rightrightarrows \prod_{\varphi : i \to j} D(j)$$

with maps

$$(a_i)_{i\in\mathcal{J}} \mapsto (a_j)_{\varphi:i\to j}$$
$$(a_i)_{i\in\mathcal{J}} \mapsto (D(\varphi)(a_i))_{\varphi:i\to j}.$$

Corollary 7.1.3 · *For any locally small category $\mathscr{C}$ we have*

$$\mathrm{Hom}_{\mathscr{C}}(X, \lim_{i\in\mathcal{J}} D(i)) \cong \lim_{i\in\mathcal{J}} \mathrm{Hom}_{\mathscr{C}}(X, D(i)),$$

$$\mathrm{Hom}_{\mathscr{C}}(\operatorname*{colim}_{i\in J} D(i), Y) \cong \lim_{i\in\mathcal{J}} \mathrm{Hom}_{\mathscr{C}}(D(i), Y).$$

Proof. (Sketch of the first isomorphism) To give a map $X \to \lim D(i)$ is to give a cone

$$(X \to D(i))_{i\in\mathcal{J}};$$

this means giving maps $\pi_i \in \mathrm{Hom}_{\mathscr{C}}(X, D(i))$ such that for all $\varphi : i \to j$ in $\mathcal{J}$ the diagram

$$\begin{tikzcd} & X \arrow[dl, "\pi_i"'] \arrow[dr, "\pi_j"] & \\ D(i) \arrow[rr, "D(\varphi)"'] & & D(j) \end{tikzcd}$$

commutes. That is, it is to give a collection $(\pi_i)_{i\in\mathcal{J}} \in \prod_{i\in\mathcal{J}} \mathrm{Hom}(X, D(i))$ such that $\mathrm{Hom}_{\mathscr{C}}(X, D(\varphi)) = \pi_j$ for all $\pi : i \to j$.

□

Corollary 7.1.4 · *A locally small category $\mathscr{C}$ has all small limits if and only if it has all small products and equalisers (the dual statement, requiring small coproducts and coequalisers, holds for colimits).*

Proof. Small products and equalisers are small limits. We thus need only prove small limits exist if small products and equalisers exist. To this end, let $D : \mathcal{J} \to \mathscr{C}$ be a diagram. Consider the equalizer

$$E = \mathrm{eq}(\prod_{i\in\mathcal{J}} D(i) \rightrightarrows \prod_{\varphi:i\to j} D(j)).$$

By Corollary 7.1.3, the set of morphisms into E can be computed as a limit in set:

$$\mathrm{Hom}_{\mathscr{C}}(X, E) \cong \mathrm{eq}(\mathrm{Hom}_{\mathscr{C}}(X, \prod_{i\in\mathcal{J}} D(i)) \rightrightarrows \mathrm{Hom}_{\mathscr{C}}(X, \prod_{\varphi:i\to j} D(j))).$$

Applying Corollary 7.1.3 again to the objects of this equalizer yields and using Remark 7.1.2 yields

$$\mathrm{Hom}_{\mathscr{C}}(X, E) \cong \mathrm{eq}(\prod_{i\in J} \mathrm{Hom}_{\mathscr{C}}(X, D(i)) \rightrightarrows \prod_{\varphi:i\to j} \mathrm{Hom}_{\mathscr{C}}(X, D(j))) \cong \lim_{i\in\mathcal{J}} \mathrm{Hom}_{\mathscr{C}}(X, D(i)).$$

□

Corollary 7.1.5 · *The colimit of any diagram in* **Set** *exists, and it is given by*

$$\mathrm{coeq}(\coprod_{\varphi:i\to j} D(i) \rightrightarrows \coprod_{i\in j} D(i))$$

Corollary 7.1.6 · *Let $D : \mathcal{J} \to \mathscr{C}$ be a diagram with $\mathcal{J}$ filtered. Then*

$$\varinjlim_{i\in\mathcal{J}} D(i) = (\coprod_{i\in\mathcal{J}} D(i))/_{\sim}$$

where $a \sim b$ for $a \in D(i)$ and $b \in D(j)$ if and only if there exist maps $\varphi : i \to k$ and $\psi : j \to k$ such that $D(\varphi)(a) = D(\psi)(b)$.

proof of this, some prose.

7.2 Limits and colimits of abelian groups

'Bootstrap, bootstrap, bootstrap.'

Lemma 7.2.1 · *Every diagram in* **Ab** *has a limit and a colimit.*

Proof. We saw (co)products and (co)equalisers in the previous lecture. Products of abelian groups are products on the underlying sets with the usual group structure. Coproducts are direct sums. Equalisers are kernels, coequalisers are cokernels:

$$\mathrm{eq}(A \underset{g}{\overset{f}{\rightrightarrows}} B) = \ker(f - g)$$
$$\mathrm{coeq}(A \underset{g}{\overset{f}{\rightrightarrows}} B) = \mathrm{coker}(f - g).$$

Thus we can use Corollary 7.1.4 to compute limits and colimits from (co)products and (co)equalizers via kernels and cokernels. □

Remark 7.2.2 · If $\mathcal{J}$ is filtered, the colimit is easier: the filtered colimit

$$\varinjlim D(i) = (\bigsqcup (i))/_\sim$$

in **Set** gets an abelian group structure as follows: if $a \in D(i)$ and $b \in D(j)$, then choose maps $\varphi : i \to k$ and $\psi : j \to k$ (using that $\mathcal{J}$ is filtered) and set

$$a + b = D(\varphi)(a) + D(\psi)(b).$$

One needs to check this is well defined and that it indeed turns the colimit in **Set** to the colimit in **Ab**. This result actually follows from a more general fact: forgetful functors create filtered colimits [Rie16, Theorem 5.6.5]

this reference deals with monadic functors, not immediately obvious it proves our claim. A small discussion of this would be nice.

7.3 Limits and colimits of presheaves

Recall that a presheaf to a category $\mathcal{D}$ is a contravariant functor $F \in \mathbf{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{D})$. In this course the category $\mathcal{D}$ has been **Set** or **Ab**, for which we know small limits and colimits exist. The following lemma appears as [Rie16, Proposition 3.3.9] stripped of the context of presheaves. It tells us that if $\mathcal{C}$ is small, the work we've done so far for **Set** and **Ab** suffices to show presheaves into these categories have small limits and colimits. Furthermore, we can compute (co)limits of such presheaves objectwise. We have seen these claims before in Lemma 2.1.3.

Lemma 7.3.1 · *Let $\mathcal{C}$ be a small category and $\mathcal{D}$ be a category with small limits and colimits. Then any diagram $D : \mathcal{J} \to \mathbf{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{D})$ has a limit and a colimit, computed objectwise:*

$$(\lim_{i \in \mathcal{J}} D(i))(U) = \lim_{i \in \mathcal{J}} (D(i)(U))$$
$$(\operatorname{colim}_{i \in \mathcal{J}} D(i))(U) = \operatorname{colim}_{i \in \mathcal{J}} (D(i)(U))$$

for all $U \in \mathcal{C}$.

Proof. We reproduce the proof given in [Rie16]. The idea is to show that $\mathbf{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{D}) \to \mathbf{Fun}(\mathcal{C}^{\mathrm{disc}}, \mathcal{D})$ □

Write the details

Corollary 7.3.2 · *The presheaf categories* $\mathbf{PSh}(X)$ *and* $\mathbf{PAb}(X)$ *have all small limits and colimits.*

7.4 Limits and colimits of sheaves

Theorem 7.4.1 · *Any diagram $D : \mathcal{J} \to \mathbf{Sh}(X)$ (or $D : \mathcal{J} \to \mathbf{Ab}(X)$) has a limit, computed in $\mathbf{PSh}(X)$ ($\mathbf{PAb}(X)$). Such a diagram also has a colimit in $\mathbf{Sh}(X)$ ($\mathbf{Ab}(X)$), obtained by sheafification of the colimit in $\mathbf{PSh}(X)$ ($\mathbf{PAb}(X)$).*

Write the rest of the lecture

LECTURE 8

Additive and abelian categories

8.1 Additive categories

'We can only write down diagrams, prove they commute, and run away as fast as we can.'

Definition 8.1.1 · A *pre-additive* category $\mathscr{C}$ is a locally small category where each hom-set $\mathrm{Hom}_{\mathscr{C}}(X, Y)$ are endowed with an abelian group structure.

Lemma 8.1.2 · *Let $\mathscr{C}$ be a pre-additive category.*

(i) If $X \in \mathscr{C}$, then the following are equivalent:
  (a) X is initial;
  (b) X is terminal;
  (c) $\mathrm{id}_X = 0 \in \mathrm{Hom}_{\mathscr{C}}(X, X)$;

(ii) If $X, Y, Z \in \mathscr{C}$ then the following are equivalent:
  (a) There are maps
$$X \xrightarrow{i} Z \xleftarrow{j} Y$$
  making Z the coproduct.
  (b) There are maps
$$Z \xrightarrow{p} X \xleftarrow{Y}$$
  making Z the product.
  (c) There are maps
$$X \underset{p}{\overset{i}{\rightleftarrows}} Z \underset{q}{\overset{j}{\leftrightarrows}} Y$$
  such that $p \circ i = \mathrm{id}_X$ and $q \circ j = \mathrm{id}_Y$ and $i \circ p + j \circ q = \mathrm{id}_Z$.

Definition 8.1.3 · A biproduct in a pre-additive category of $X, Y \in \mathscr{C}$ is an object Z with maps
$$X \underset{p}{\overset{i}{\rightleftarrows}} Z \underset{q}{\overset{j}{\leftrightarrows}} Y$$
with $pi = \mathrm{id}_X$, $qj = \mathrm{id}_Y$ and $ip + jq = \mathrm{id}_Z$.

Remark 8.1.4 · Note that these force $q \circ i$ and $p \circ j$ to be zero. To see this it suffices to show that $j \circ q \circ i = 0$, because j is monic. This holds because $jqi = (\mathrm{id}_Z - ip)i = i - ipi = i - i = 0$.

Proof. (Of the lemma)

(i) If x is initial, then $\mathrm{Hom}_{\mathscr{C}}(X, X) = 0$ so $\mathrm{id}_X = 0$. Conversely, if $\mathrm{id}_x = 0$ then every map $f : X \to Y$ i n $\mathscr{C}$ satisfies $f = f \circ \mathrm{id}_X = f \circ 0 = 0$. So X is initial. This proves (i) is equivalent to (ii) and that (ii) is equivalent to (iii) follows dually.

(ii) If

$$X \underset{p}{\overset{i}{\rightleftarrows}} Z \underset{q}{\overset{j}{\leftrightarrows}} Y$$

is a biproduct, we saw $qi = 0$ and $pj = 0$. Then $X \xrightarrow{i} Z$ and $Y \xrightarrow{j} Z$ is a coproduct. If

$$X \xrightarrow{f} W \xleftarrow{g} Y$$

is any cocone, set $h = fp + gq : Z \to W$. Then $hi = (fp + gq)i = fpi + gqi = f \circ \mathrm{id}_X + g \circ 0 = f$, and likewise $hj = g$. If $h' : Z \to W$ satisfies $h'i = f$ and $h'j = g$ then $h' = h' \circ \mathrm{id}_Z = h' \circ (ip + jq) = h'ip + h'jq = fp + gq = h$. So

$$X \xrightarrow{i} Z \xleftarrow{j} Y$$

form a coproduct. Conversely, if these maps form a coproduct, then the cocones

$$X \xrightarrow{\mathrm{id}} X \xleftarrow{0} Y$$
$$X \xrightarrow{0} Y \xleftarrow{\mathrm{id}} Y$$

define maps $p : Z \to X$, $q : Z \to Y$ with $pi = \mathrm{id}_X, pj = 0, qi = 0, qj = \mathrm{id}_Y$. Then $ip + jq : Z \to Z$ satisfies $(ip + jq)i = ipi + jqi = i \circ \mathrm{id}_X + j \circ 0 = i$, and likewise $(ip + jq) \circ j = \cdots = j$. So $ip + jq = \mathrm{id}_Z$ by the universal property of the coproduct. This proves $(i) \Leftrightarrow (iii)$ and $(ii) \Leftrightarrow$ follows dually.

□

Definition 8.1.5 · An *additive* category is a pre-additive category with finite products.

Remark 8.1.6 · By the lemma Lemma 8.1.2 if $\mathscr{C}$ is additive then

- the terminal object is also initial, hence a zero object 0.
- IF $X, Y \in \mathscr{C}$ then the biproduct

$$X \underset{p}{\overset{i}{\rightleftarrows}} X \oplus Y \underset{q}{\overset{j}{\leftrightarrows}} Y$$

  exists, so $\mathscr{C}$ has finite coproducts.
- If $X_1, \cdots, X_n \in \mathscr{C}$ then the map $X_1 \coprod \cdots \coprod X_n \to X_1 \times \cdots \times X_n$ such that $X_i \to X_1 \coprod \cdots \coprod X_n \to X_1 \times \cdots \times X_n \to X_j$ is id_X if $i = j$ and 0 if $i \neq j$ is an isomorphism. We write $X_1 \oplus \cdots \oplus X_n$ for the n-ary biproduct.

Definition 8.1.7 · A *semi-additive* category is a category $\mathscr{C}$ with finite products and coproducts such that

(i) the natural map $\varnothing \to *$ is an isomorphism (so $\mathscr{C}$ has a zero object 0),
(ii) for $X_1, \ldots, X_n$ in $\mathscr{C}$ the map $X_1 \coprod \cdots \coprod X_n \to X_1 \times \cdots \times X_n$ is an isomorphism.

(Note that we use (i) to define the map in (ii)).

Remark 8.1.8 · While (pre)-additive is a *structure*, being semi-additive is a *property* of a category $\mathscr{C}$.

Lemma 8.1.9 · *If $\mathscr{C}$ is semi-additive, then it is canonically enriched in commutative monoids. If $\mathscr{C}$ was additive, then this agrees with the given enrichment in abelian groups (under the inclusion* **Ab** $\hookrightarrow$ **CMon***).*

Proof. (Sketch, in which the commutativity of the many diagrams is left to the reader,) For $f, g : X \rightrightarrows Y$ in $\mathscr{C}$, define $f + g$ to be

$$X \xrightarrow{\Delta} X \times X = X \oplus X \xrightarrow{f \oplus g} Y \oplus Y = Y \coprod Y \xrightarrow{\nabla} Y$$

where Δ is the diagonal map and ∇ is the codiagonal. For f, g, h maps from X to Y, commutativity of

$$\begin{tikzcd}
 & X \oplus X \arrow[rr, "(f+g)\oplus h"] \arrow[d, "\Delta\oplus \mathrm{id}_X"] & & Y \oplus Y \arrow[dr, "\Delta"] & \\
X \arrow[ur, "\Delta"] \arrow[r, "\Delta"] \arrow[dr, "\Delta"'] & X \oplus X \oplus X \arrow[rr, "f\oplus g\oplus h"] & & Y \oplus Y \oplus Y \arrow[u, "\nabla\oplus \mathrm{id}"] \arrow[d, "\mathrm{id}_Y\oplus\nabla"] & Y \\
 & X \oplus X \arrow[u, "\mathrm{id}_X\oplus\Delta"] \arrow[rr, "f\oplus(g+h)"'] & & Y \oplus Y \arrow[ur, "\nabla"'] &
\end{tikzcd}$$

shows that $(f+g)+h = f+(g+h)$. For $f : X \to Y$ commutativity of

$$\begin{tikzcd}
 & X \oplus X \arrow[r, "f\oplus 0"] \arrow[d, "\mathrm{id}\oplus 0"] & Y \oplus Y \arrow[dr, "\nabla"] & \\
X \arrow[ur, "\Delta"] \arrow[r] & X \oplus 0 \arrow[r, "f\oplus\mathrm{id}"'] & Y \oplus 0 \arrow[u, "\mathrm{id}\oplus 0"] \arrow[r] & Y
\end{tikzcd}$$

shows $f + 0 = f$ and likewise $0 + f = f$. For $f, g : X \rightrightarrows Y$ commutativity of

$$\begin{tikzcd}
 & X \oplus X \arrow[r, "f\oplus g"] \arrow[d, "\mathrm{swap}"] & Y \oplus Y \arrow[dr, "\nabla"] \arrow[d, "\mathrm{swap}"] & \\
X \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & & & Y \\
 & X \oplus X \arrow[r, "g\oplus f"'] & Y \oplus Y \arrow[ur, "\nabla"'] &
\end{tikzcd}$$

gives $f + g = g + f$ so have $\mathrm{Hom}_{\mathscr{C}}(X, Y)$ is a commutative monoid. If $f, f' : X \rightrightarrows Y$ and $g, g' : Y \rightrightarrows Z$ then commutativity of

$$\begin{tikzcd}
X \arrow[r, "\Delta"] & X \oplus X \arrow[r, "f\oplus f'"] \arrow[dr, "gf\oplus gf'"'] & Y \oplus Y \arrow[r, "\nabla"] \arrow[d] & Y \arrow[d, "g"] \\
 & & Z \oplus Z \arrow[r, "\nabla"] & Z
\end{tikzcd}$$

shows that $g(f + f') = gf + gf'$ and likewise $(g + g')f = gf + g'f$, proving the first statement. If $\mathscr{C}$ is additive and $f, g : X \rightrightarrows Y$ then let

$$X \underset{p_1}{\overset{i_1}{\rightleftarrows}} X \oplus X \underset{p_2}{\overset{i_2}{\leftrightarrows}} X$$

be the biproduct, and likewise for Y. Then commutativity of

$$\begin{tikzcd}
X \arrow[r, "p_1"] \arrow[d, "f"] & X \oplus X \arrow[r, "p_2"] \arrow[d, dashed, "f\oplus g"] & X \arrow[d, "g"] \\
Y & Y \oplus Y \arrow[l] \arrow[r, "p_2"'] & Y
\end{tikzcd}
\qquad
\begin{tikzcd}
X \arrow[r, "i_1"] \arrow[d, "f"'] & X \oplus X \arrow[d, "f\oplus g"] & X \arrow[l, "i_2"'] \arrow[d, "g"] \\
Y \arrow[r, "i_1"'] & Y \oplus Y & Y \arrow[l, "i_2"]
\end{tikzcd}$$

gives $f \oplus g = (f \oplus g) \circ \mathrm{id}_{X\oplus X} = (f \oplus g) \circ (i_1 p_1 + i_2 p_2) = i_1 f p_1 + i_2 g p_2$, and likewise $f \oplus 0 = i_1 f p_1$, $0 \oplus g = i_2 g p_2$, so

$$f \oplus g = (f \oplus 0) + (0 \oplus g).$$

Then bilinearity of composition shows that the composite

$$X \xrightarrow{\Delta} X \oplus X \xrightarrow{f\oplus g} Y \oplus Y \nabla Y$$

is $f + g$ (in the enriched sense). $\square$

Corollary 8.1.10 · *Being additive (i.e. admitting an additive structure) is a property.*

Proof. The category $\mathscr{C}$ is additive if and only if if it is semi-additive and all Hom monoids are groups. □

Example 8.1.11 · **CMon** is semi-additive (omitted), but not additive ($\mathrm{Hom}(\mathbb{N}, \mathbb{N}) = \mathbb{N}$ does not have inverses).

8.2 Abelian categories

Exercise 8.2.1 · If $\mathscr{C}$ is pre-additive, show that

$$\mathrm{eq}(X \rightrightarrows f[g] = \mathrm{eq}(X \rightrightarrows (f-g)[0]Y)) =: \ker(f-g)$$

(if one of them exists).

So an additive category $\mathscr{C}$ has finite limits (respectively colimits) if and only if it has kernels (respectively cokernels).

Definition 8.2.2 · A *pre-abelian* category is an additive category with kernels and cokernels.

Definition 8.2.3 · Let $\mathscr{C}$ be a pre-abelian category and $f : X \to Y$ in $\mathscr{C}$. The *image* of f is $\mathrm{im}(f) = \ker(Y \to \mathrm{coker}(f))$. The *coimage* of f is $\mathrm{coim}(f) = \mathrm{coker}(\ker(f) \to X)$.

Warning: the zoo of names will get worse.

Lemma 8.2.4 · *Any $f : X \to Y$ in a pre-abelian category factors uniquely via*

$$X \to \mathrm{coim}(f) \to \mathrm{im}(f) \to Y.$$

Proof. The composition $\ker(f) \to X \to Y$ is 0. So there is a unique factorisation

$$X \to \mathrm{coim}(f) \to Y$$

of f. The composition $\mathrm{coim}(f) \to Y \to \mathrm{coker}(f)$ is 0 (since this holds after precomposition with $X \to \mathrm{coim}(f)$). So there is a unique factorisation

$$\mathrm{coim}(f) \to \mathrm{im}(f) \to Y.$$

□

And now the moment we have all been waiting for: the definition of an abelian category.

Definition 8.2.5 · An *abelian* category is a pre-abelian category A such that the $\mathrm{coim}(f) \to \mathrm{im}(f)$ is an isomorphism for all $f : X \to Y$ in $\mathscr{A}$.

Example 8.2.6 · In **Ab**, $\mathbf{Mod}_R$ this is the first isomorphism theorem: $X/\ker(f) \xrightarrow{\sim} \mathrm{im}(f)$.

Lemma 8.2.7 · *If $\mathscr{A}$ is abelian and $\mathscr{C}$ is small, then* $\mathbf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathscr{A})$ *is abelian.*

This lemma implies that the category of presheaves of abelian groups on a topological space is an abelian category.

Proof. If $\alpha, \beta : F \implies G$ is a natural transformation, then define the sum $\alpha + \beta : F \implies G$ by $(\alpha + \beta)_U = \alpha_U + \beta_U$. This is natural since the maps $F(V) \to F(U)$ and $G(V) \to G(U)$ are group homorphisms for any map $U \to V$ in $\mathscr{C}$. This turns $\mathbf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathscr{A})$ into a pre-additive category. All the other questions (existence of finite, limits and colimits, the first isomorphism theroem) are checked objectwise. □

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